pfSense Firewall Deployment & Security Configuration

Blue Team Internship

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Institute: ITSimplera Institute

1. Introduction to pfSense

pfSense is a free, open-source firewall and router software distribution based on FreeBSD. It is widely adopted in both enterprise and home-lab environments because it transforms standard hardware or a virtual machine into a full-featured, enterprise-grade security appliance without licensing costs. pfSense combines the core functions of a stateful packet-filtering firewall with a rich set of networking and security services, all managed through a clean, browser-based Web Configurator (WebGUI) rather than a command-line-only interface.

At its core, pfSense provides stateful firewalling, Network Address Translation (NAT), and traffic shaping, built on the mature and highly reliable pf (packet filter) engine inherited from FreeBSD/OpenBSD. Beyond basic filtering, it supports site-to-site and remote-access VPNs (IPsec, OpenVPN, WireGuard), load balancing and failover across multiple WAN connections, DHCP and DNS services (including DNS Resolver/Unbound), and detailed traffic monitoring and reporting.

One of pfSense's biggest strengths is its extensible package system. Through System > Package Manager, administrators can add specialized functionality on top of the base firewall, such as:

- pfBlockerNG — GeoIP-based country blocking and DNSBL (DNS-layer) content/advertising/malware filtering.
- Suricata / Snort — signature-based Intrusion Detection and Prevention (IDS/IPS).
- HAProxy — advanced load balancing and reverse proxying.
- OpenVPN Client Export / Squid — VPN client provisioning and web proxy/caching services.

Because of this flexibility, pfSense is commonly used as a perimeter firewall, a segmentation boundary between network zones (e.g. LAN, DMZ, guest Wi-Fi), or a VPN concentrator for remote access. For a SOC or Blue Team learner, pfSense is an ideal training platform: it exposes the same fundamental concepts used in commercial firewalls — access control rules, NAT, logging, alerting, and threat-intelligence-driven blocking — in an environment that is free to deploy, easy to reset, and safe to experiment with in an isolated lab.

In this lab, pfSense serves as the security gateway for a small simulated network. On top of the base firewall, the pfBlockerNG package is installed to add two specific threat-mitigation capabilities: GeoIP-based country blocking (denying traffic to/from selected high-risk countries) and DNSBL-based website filtering (blocking access to specified domains at the DNS layer). Administrative access to the WebGUI is then hardened, and finally, firewall logs and pfBlockerNG reports are used to monitor and validate that every configured policy is functioning correctly — mirroring the day-to-day responsibilities of a SOC analyst maintaining a network perimeter.

2. Objective

As a SOC Intern at ITSimplera Institute, the objective of this lab is to deploy and configure the pfSense firewall to establish a secure and monitored network perimeter. This includes installing pfSense as a virtual appliance, hardening baseline firewall settings, installing and configuring the pfBlockerNG package, enforcing GeoIP-based country blocking, filtering malicious or unwanted websites using DNSBL, restricting administrative access to the WebGUI, and finally monitoring firewall logs and pfBlockerNG reports to validate that all security policies are functioning as intended.

This exercise builds foundational skills in firewall administration, network traffic filtering, and security event monitoring — core competencies for Blue Team and Network Security operations.

4.Installation

Network Configuration for pfSense in VMware

pfSense requires multiple network interfaces because it acts as both a firewall and a router.

Network Interfaces

- WAN Interface → VMnet8 (NAT)
    - Provides Internet connectivity to the pfSense firewall.
- LAN Interface → VMnet2 (Host-Only)
    - Connects the internal network (e.g., Windows and Kali virtual machines) to the pfSense firewall.

Disable DHCP on VMnet2

The DHCP service on VMnet2 must be disabled in VMware's Virtual Network Editor. This ensures that pfSense is the only DHCP server on the LAN.

If DHCP remains enabled on VMnet2, there will be two DHCP servers on the same network:

- VMware DHCP Server
- pfSense DHCP Server

This can lead to several issues:

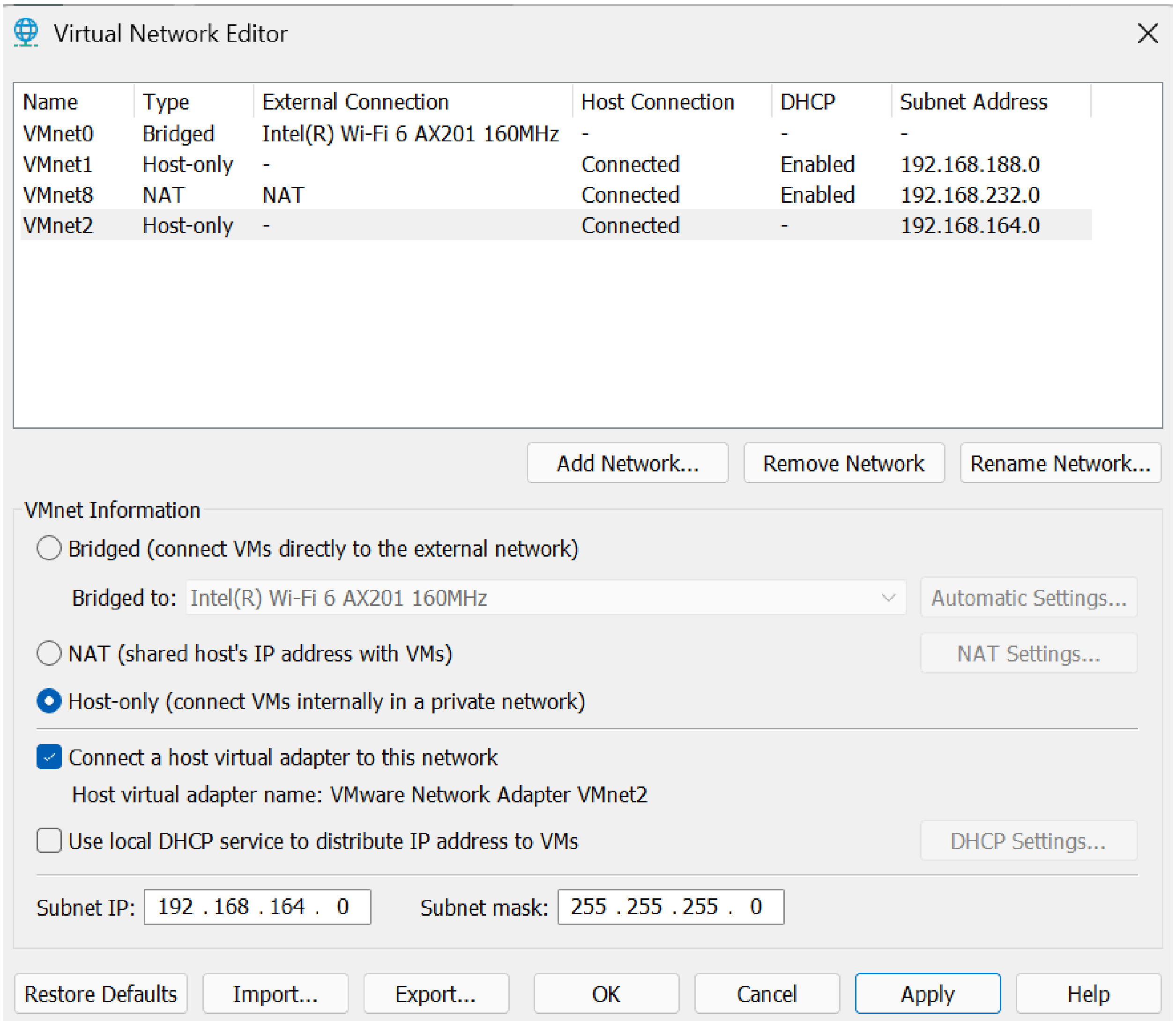
- One device may receive its IP address from VMware DHCP.
- Another device may receive its IP address from pfSense DHCP.
- Devices may be assigned different default gateways.
- Devices may use different DNS servers.
- Some traffic may bypass the pfSense firewall entirely.

As a result, the network may experience:

- IP address conflicts
- Inconsistent routing
- DNS resolution issues
- Unstable network behavior
- Firewall and pfBlockerNG policies not being applied correctly

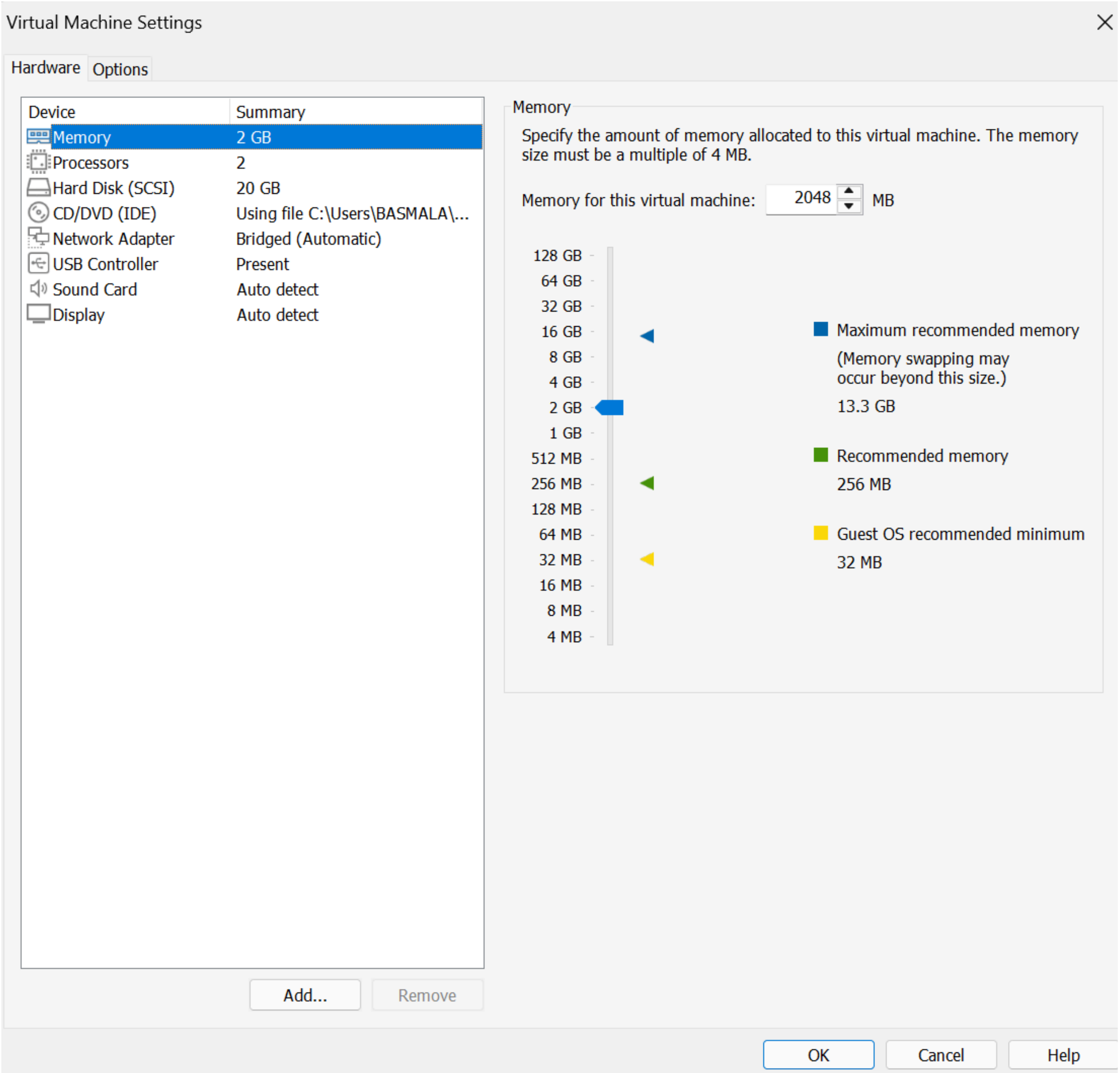
Best Practice

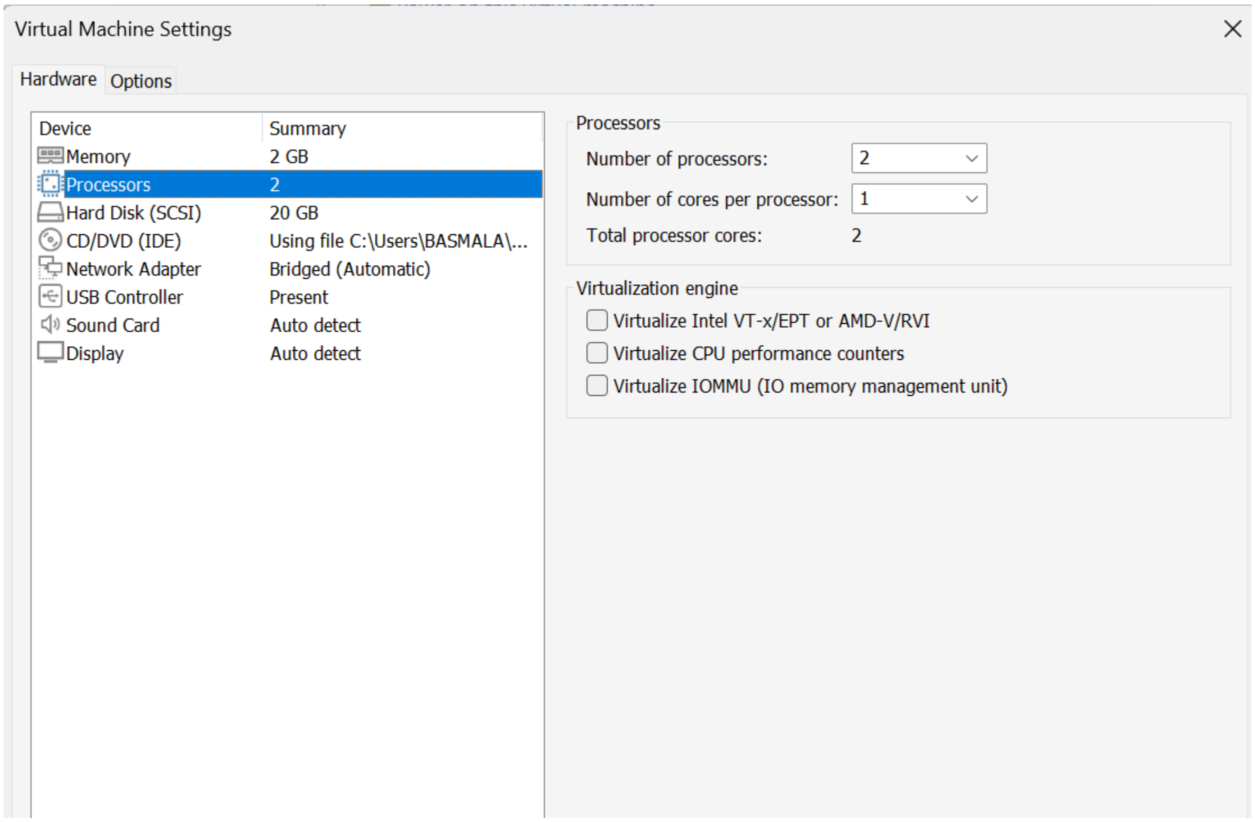
Disable VMware DHCP on VMnet2 and enable the DHCP Server on the LAN interface of pfSense. This ensures that all client devices receive their IP configuration (IP address, gateway, and DNS server) from pfSense, allowing all traffic to pass through the firewall and enabling consistent enforcement of firewall rules, GeoIP blocking, and DNSBL filtering.

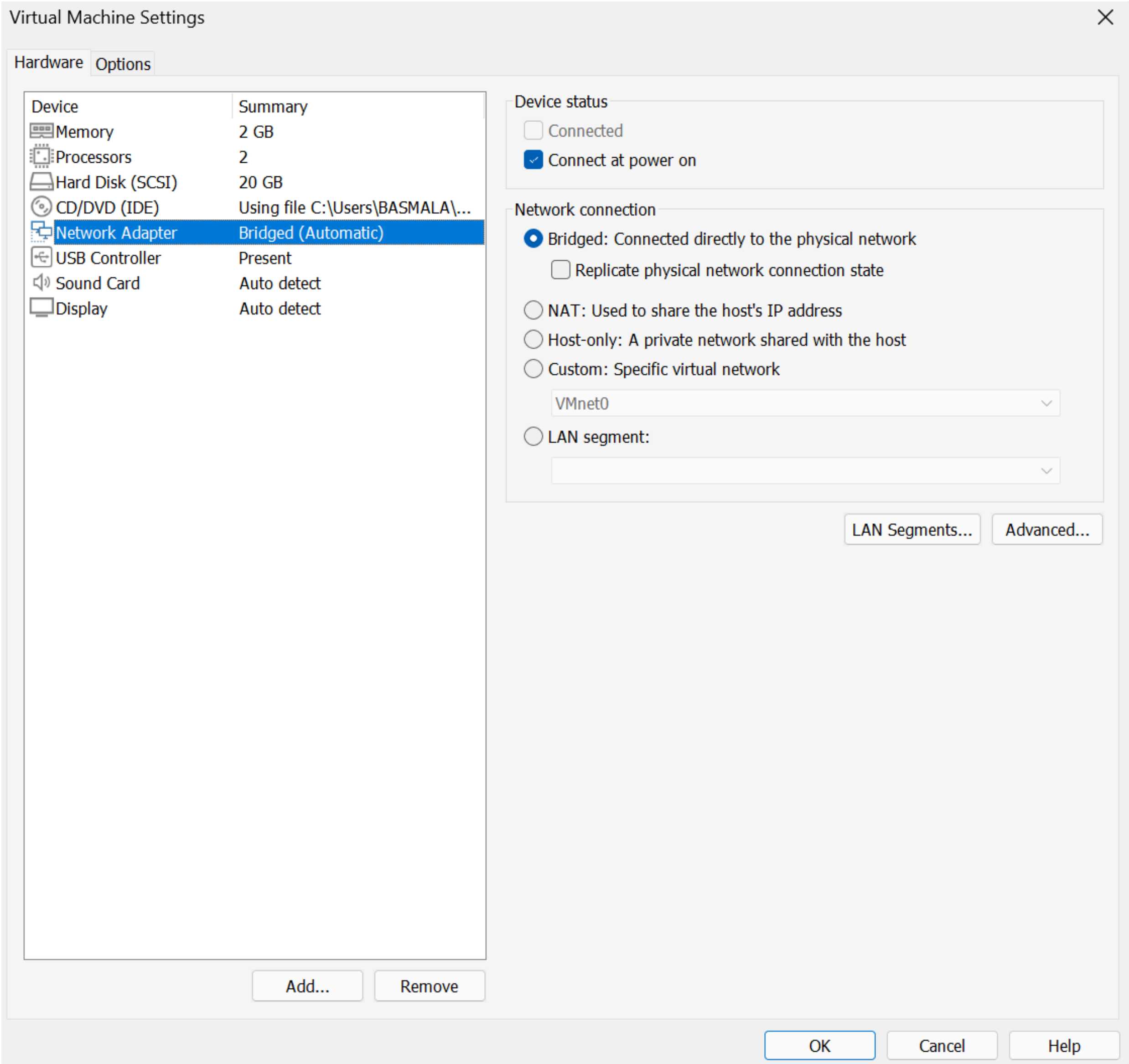


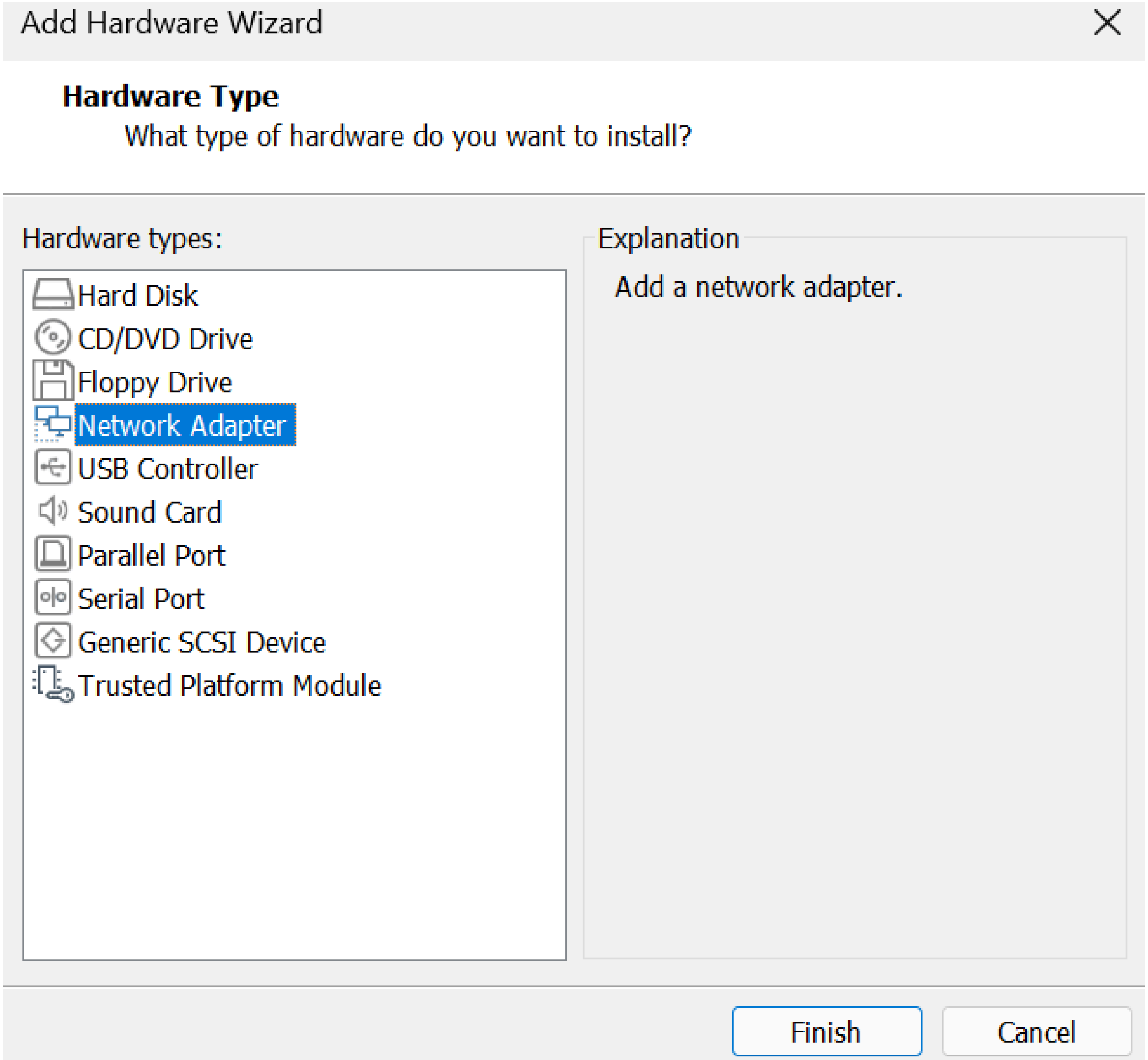
Deploy the pfSense Virtual Appliance

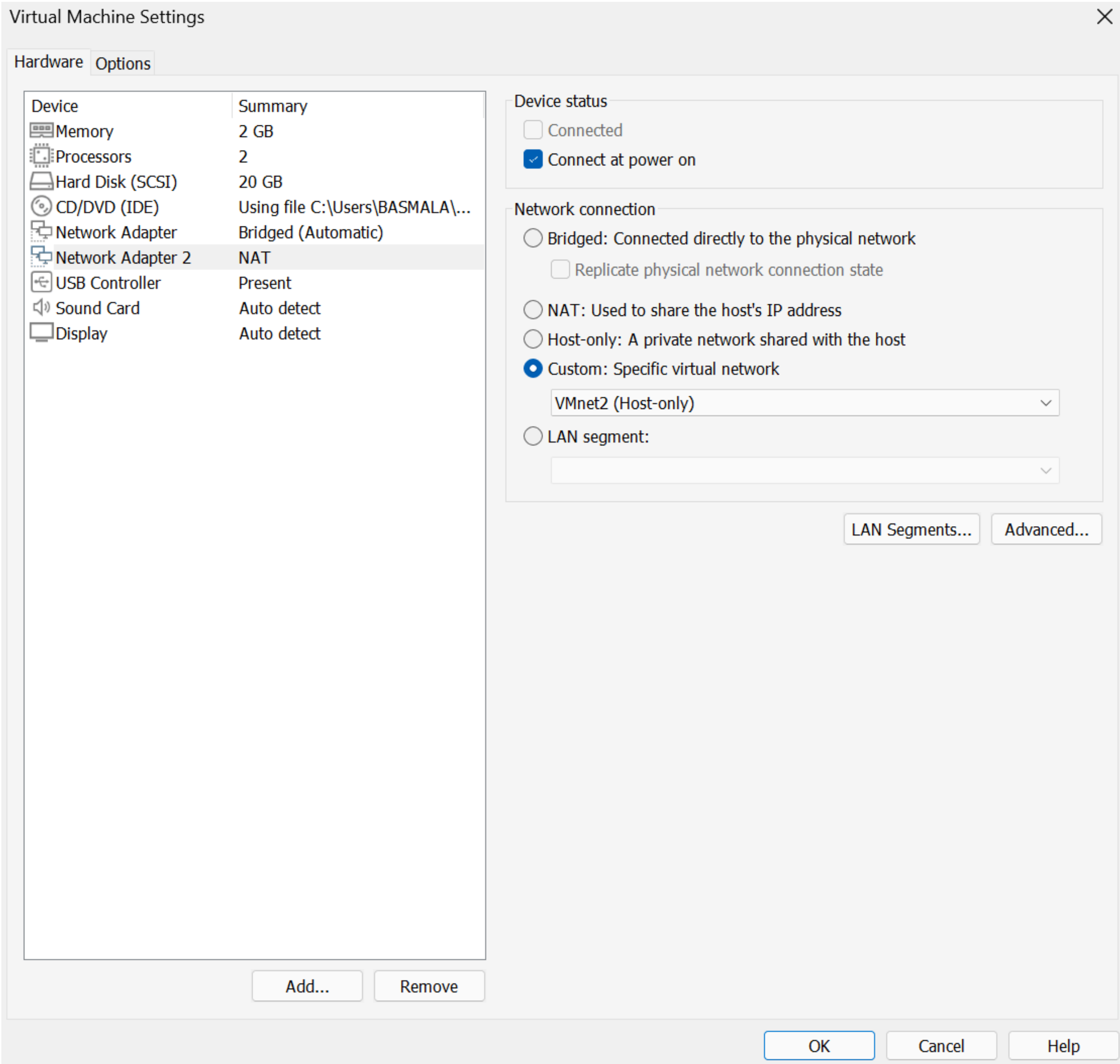
1. Download the pfSense CE ISO image from the official Netgate site.
2. Create a new virtual machine with at least 2 vCPUs, 2 GB RAM, 20 GB disk, and two virtual network adapters (WAN and LAN).
3. Attach the pfSense ISO to the VM's optical drive and boot the VM.
4. Follow the installer prompts: accept the copyright/license notice, choose 'Install pfSense', select the default keymap, and use Auto (ZFS or UFS) partitioning.
5. Allow the installation to complete, remove the ISO, and reboot into pfSense.
6. At the console menu, assign WAN and LAN interfaces to the correct virtual NICs (option 1), then configure the LAN interface IP address (e.g. 192.168.1.1/24) with DHCP server enabled (option 2).


Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
2 GB
Processors
2
Hard Disk (SCSI)
20 GB
CD/DVD (IDE)
Using file C:\Users\BASMALA\...
Network Adapter
Bridged (Automatic)
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Add...
Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine:
2048
MB
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.3 GB
Recommended memory
256 MB
Guest OS recommended minimum
32 MB
OK
Cancel
Help


Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
2 GB
Processors
2
Hard Disk (SCSI)
20 GB
CD/DVD (IDE)
Using file C:\Users\BASMALA\...
Network Adapter
Bridged (Automatic)
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Processors
Number of processors:
2
Number of cores per processor:
1
Total processor cores:
2
Virtualization engine
Virtualize Intel VT-x/EPT or AMD-V/RVI
Virtualize CPU performance counters
Virtualize IOMMU (IO memory management unit)


Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
2 GB
Processors
2
Hard Disk (SCSI)
20 GB
CD/DVD (IDE)
Using file C:\Users\BASMALA\...
Network Adapter
Bridged (Automatic)
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Add...
Remove
Device status
Connected
Connect at power on
Network connection
Bridged: Connected directly to the physical network
Replicate physical network connection state
NAT: Used to share the host's IP address
Host-only: A private network shared with the host
Custom: Specific virtual network
VMnet0
LAN segment:
LAN Segments...
Advanced...
OK
Cancel
Help


Add Hardware Wizard
Hardware Type
What type of hardware do you want to install?
Hardware types:
Hard Disk
CD/DVD Drive
Floppy Drive
Network Adapter
USB Controller
Sound Card
Parallel Port
Serial Port
Generic SCSI Device
Trusted Platform Module
Explanation
Add a network adapter.
Finish
Cancel


Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
2 GB
Processors
2
Hard Disk (SCSI)
20 GB
CD/DVD (IDE)
Using file C:\Users\BASMALA\...
Network Adapter
Bridged (Automatic)
Network Adapter 2
NAT
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Add...
Remove
Device status
Connected
Connect at power on
Network connection
Bridged: Connected directly to the physical network
Replicate physical network connection state
NAT: Used to share the host's IP address
Host-only: A private network shared with the host
Custom: Specific virtual network
VMnet2 (Host-only)
LAN segment:
LAN Segments...
Advanced...
OK
Cancel
Help

```
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.9/24
 LAN (lan) -> em1 -> v4: 192.168.1.4/24

 0) Logout / Disconnect SSH            9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart GUI
 3) Reset admin account and password  12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option:
```


```
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.9/24
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 8) Shell

Enter an option:
```



```
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

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 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option:
```
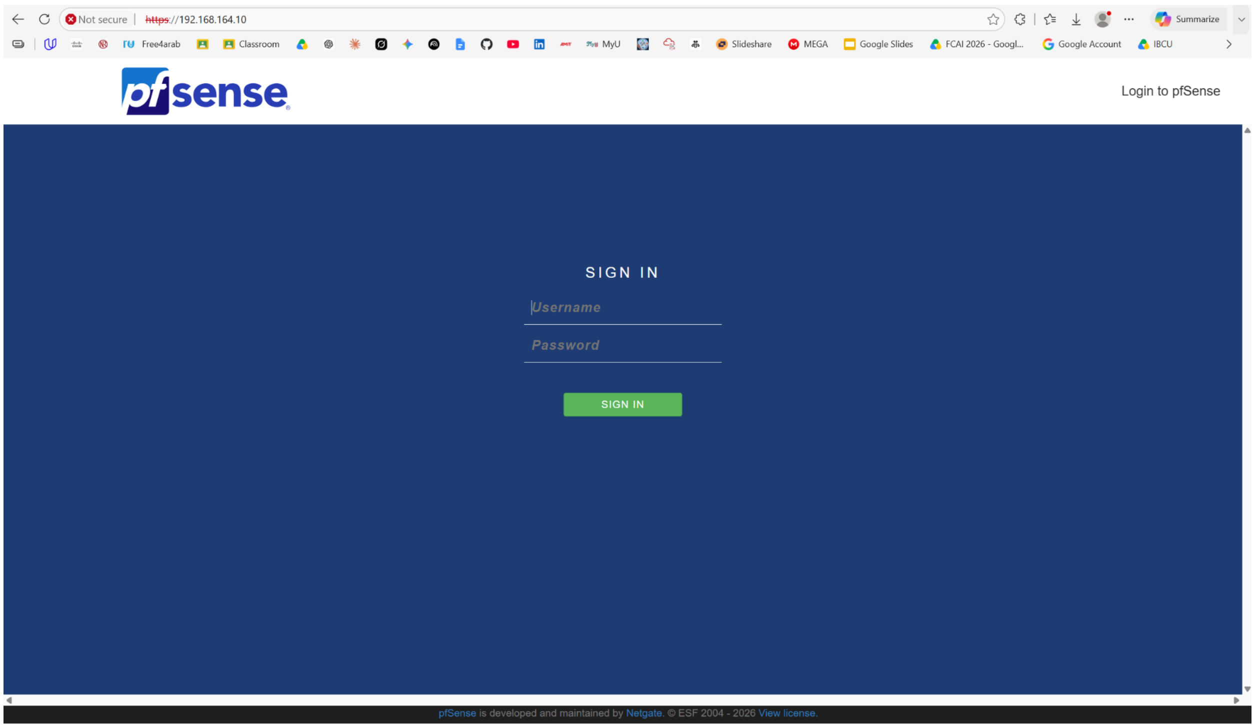
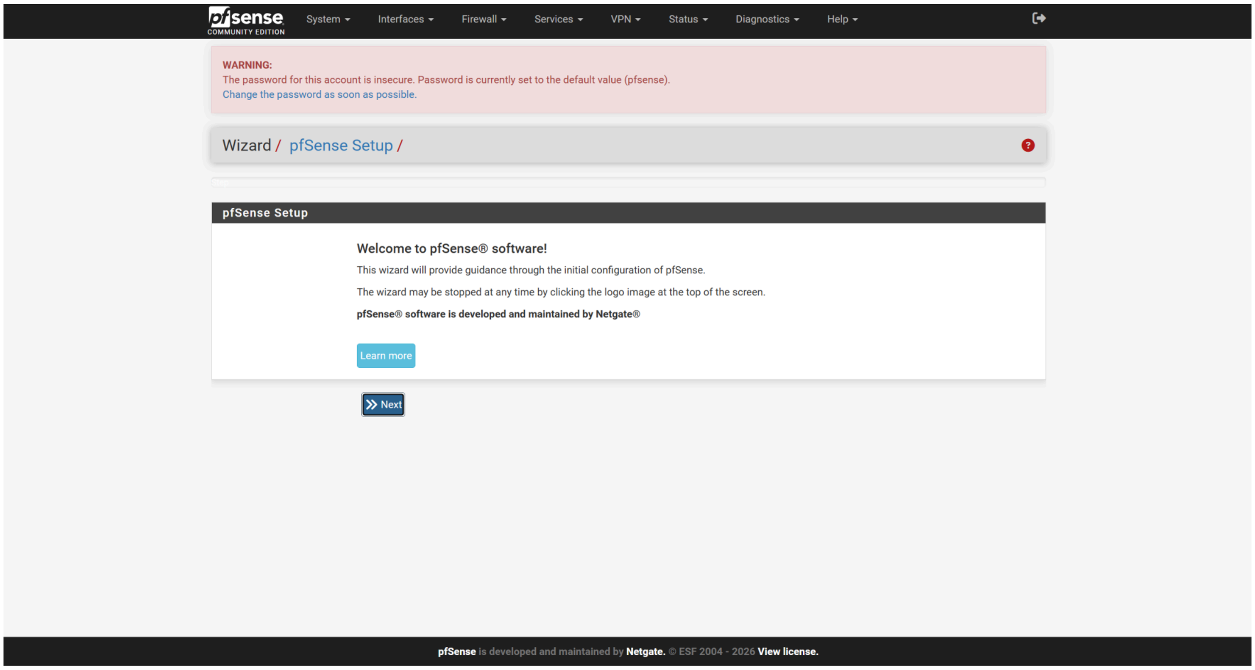
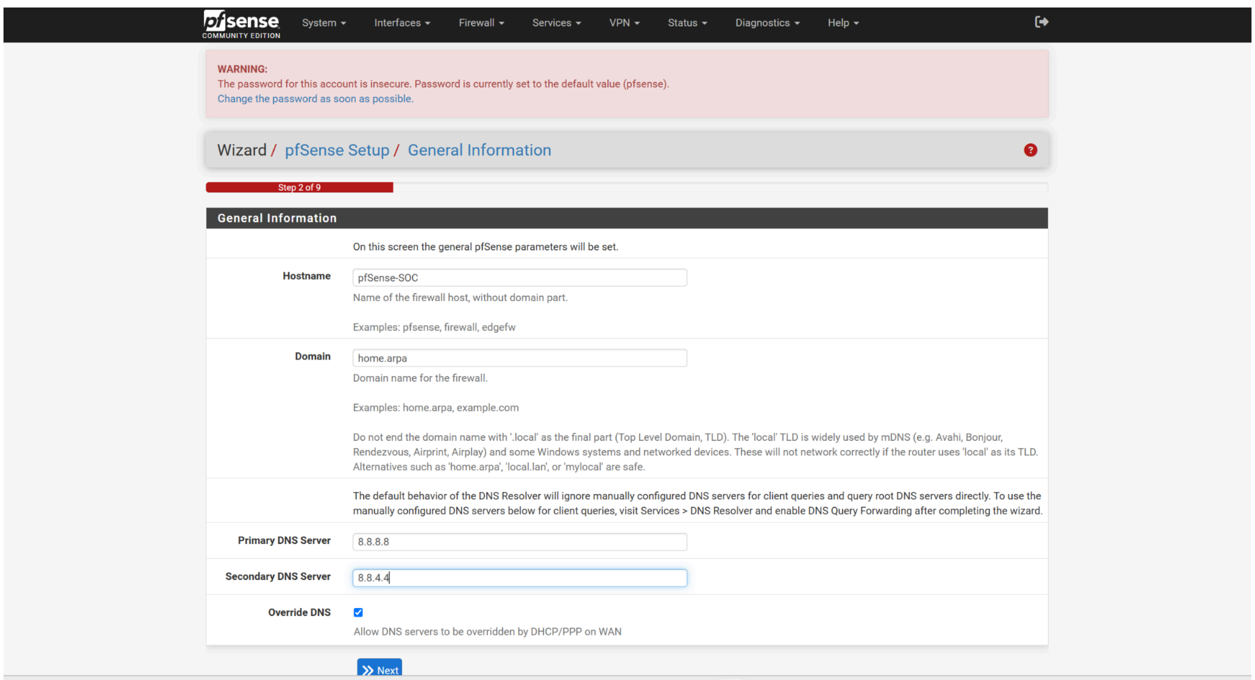

Initial pfSense Configuration & WebGUI Access

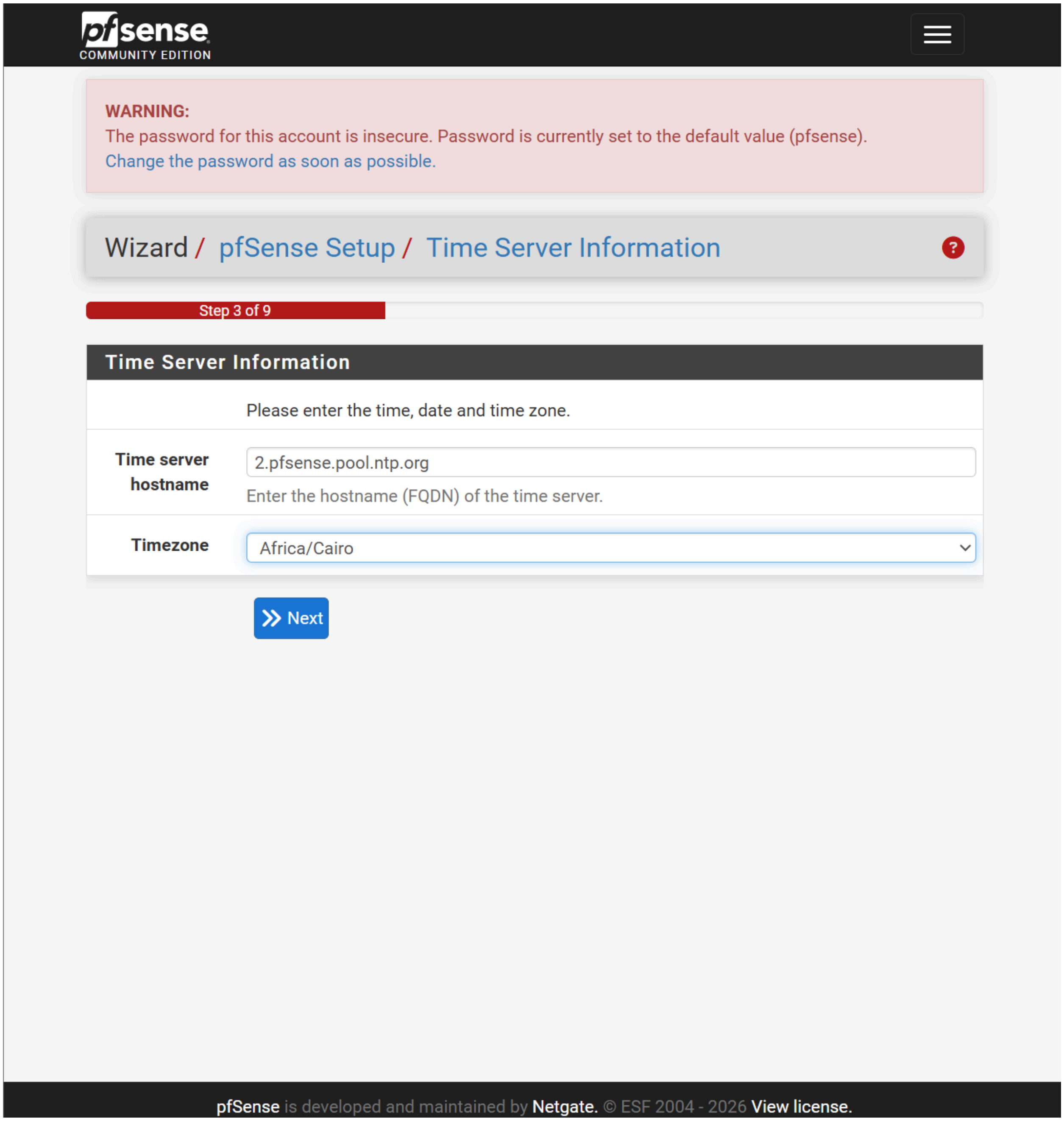
1. From the admin workstation on the LAN segment, browse to https://192.168.1.1 (or the configured LAN IP).
2. Accept the self-signed certificate warning and log in with the default credentials (admin / pfsense).
3. Complete the Setup Wizard: set the hostname, domain, DNS servers, timezone, and change the default administrator password to a strong password immediately.
4. Confirm the WAN and LAN interface configuration on the Interfaces menu and verify connectivity (WAN shows a valid IP; LAN clients receive DHCP leases).
5. Navigate to Status > Dashboard to confirm both interfaces are up and the system is healthy.



User:admin
Password:pfsense


pfsense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.
Wizard / pfSense Setup /
pfSense Setup
Welcome to pfSense® software!
This wizard will provide guidance through the initial configuration of pfSense.
The wizard may be stopped at any time by clicking the logo image at the top of the screen.
pfSense® software is developed and maintained by Netgate®
Learn more
Next
pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.


pfsense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.
Wizard / pfSense Setup / General Information
Step 2 of 9
General Information
On this screen the general pfSense parameters will be set.
Hostname
pfSense-SOC
Name of the firewall host, without domain part.
Examples: pfsense, firewall, edgefw
Domain
home.arpa
Domain name for the firewall.
Examples: home.arpa, example.com
Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.
The default behavior of the DNS Resolver will ignore manually configured DNS servers for client queries and query root DNS servers directly. To use the manually configured DNS servers below for client queries, visit Services > DNS Resolver and enable DNS Query Forwarding after completing the wizard.
Primary DNS Server
8.8.8.8
Secondary DNS Server
8.8.4.4
Override DNS
Allow DNS servers to be overridden by DHCP/PPP on WAN
Next


pfsense
COMMUNITY EDITION
WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.
Wizard / pfSense Setup / Time Server Information
Step 3 of 9
Time Server Information
Please enter the time, date and time zone.
Time server hostname
2.pfsense.pool.ntp.org
Enter the hostname (FQDN) of the time server.
Timezone
Africa/Cairo
Next
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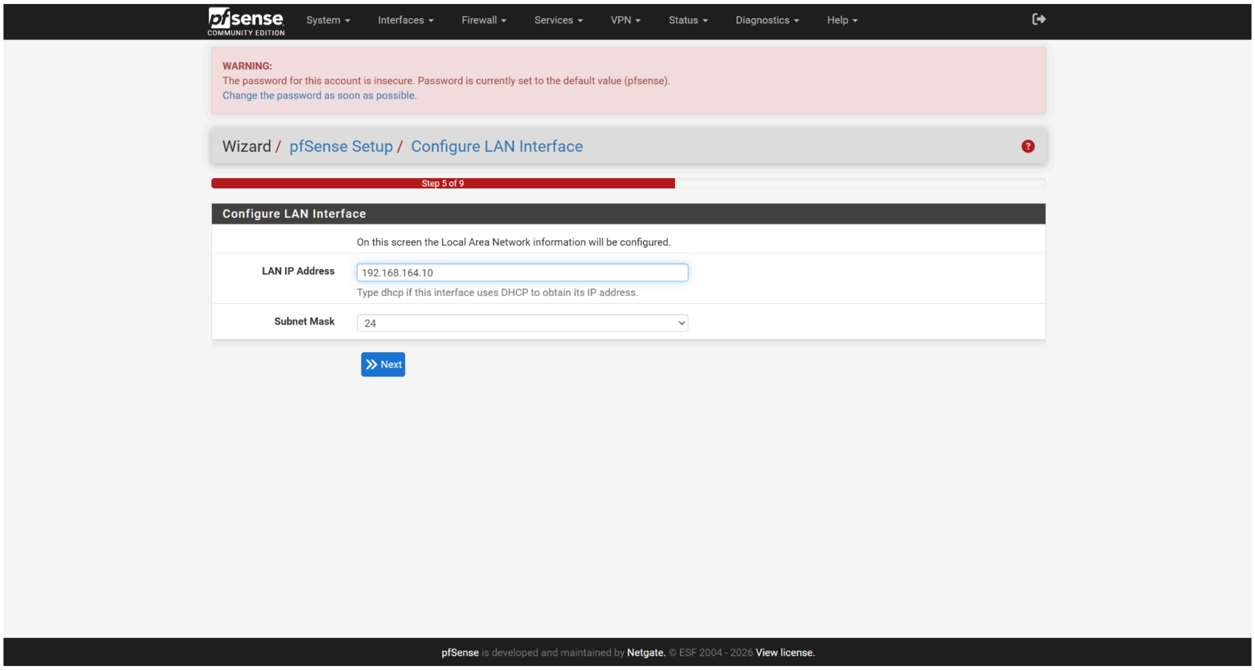
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.
Wizard / pfSense Setup / Configure LAN Interface
Step 5 of 9
Configure LAN Interface
On this screen the Local Area Network information will be configured.
LAN IP Address
192.168.164.10
Type dhcp if this interface uses DHCP to obtain its IP address.
Subnet Mask
24
Next
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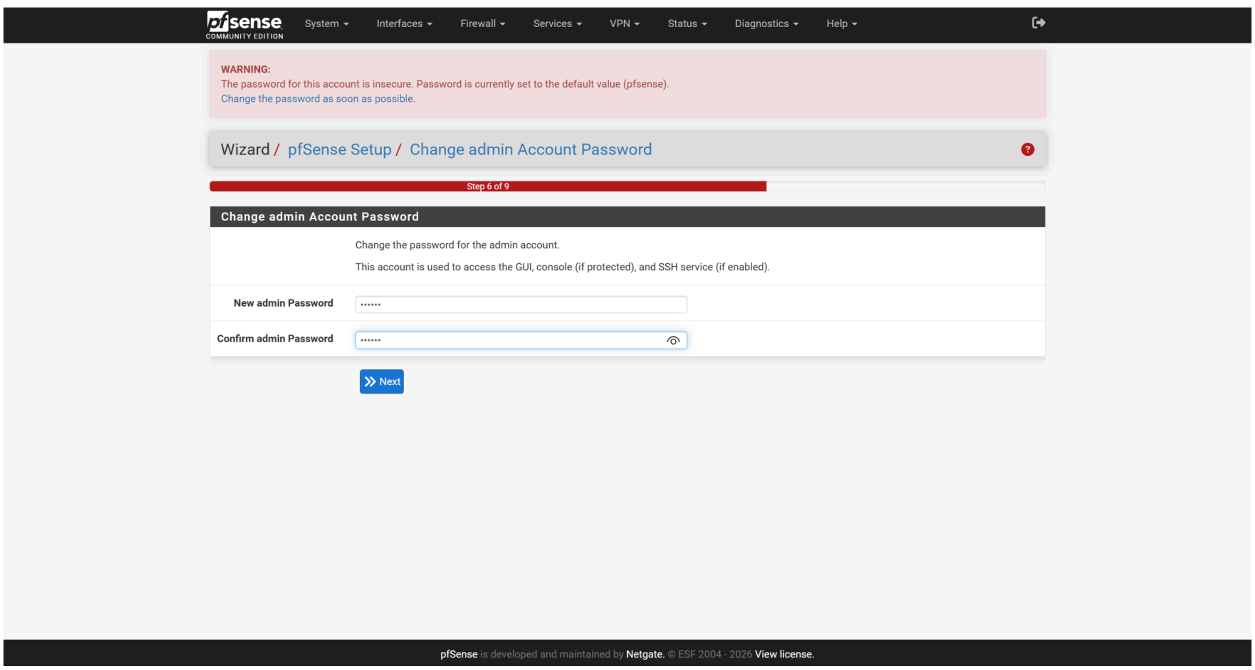
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.
Wizard / pfSense Setup / Change admin Account Password
Step 6 of 9
Change admin Account Password
Change the password for the admin account.
This account is used to access the GUI, console (if protected), and SSH service (if enabled).
New admin Password
Confirm admin Password
Next
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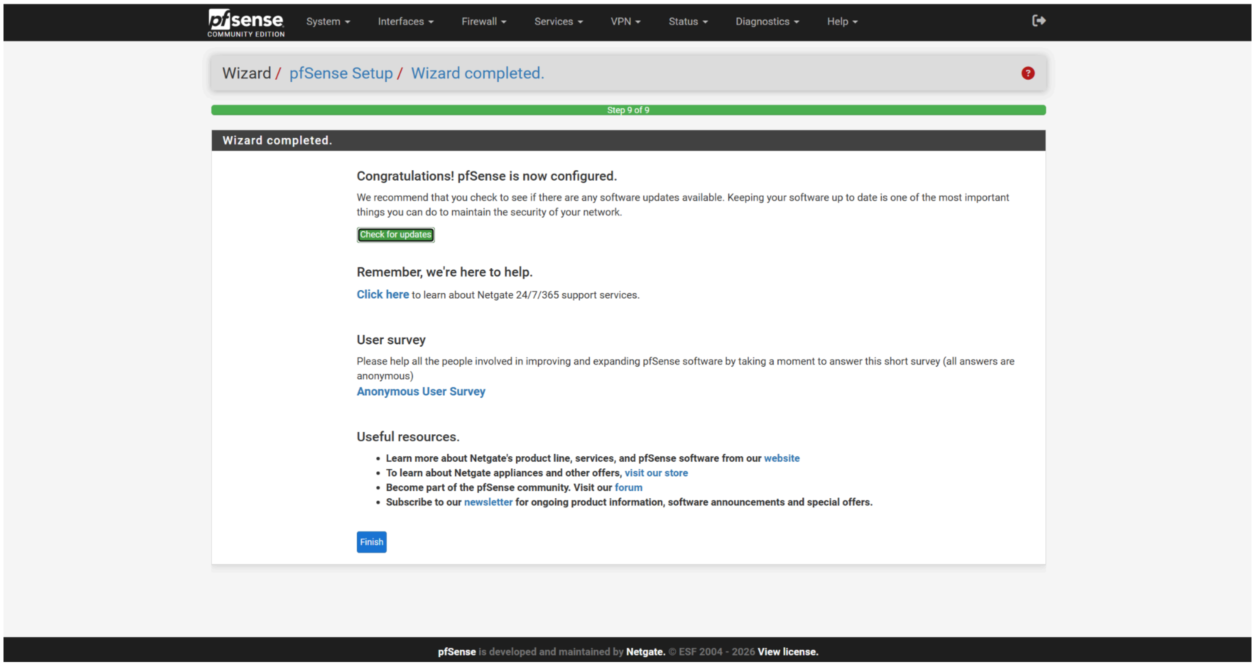
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Wizard / pfSense Setup / Wizard completed.
Step 9 of 9
Wizard completed.
Congratulations! pfSense is now configured.
We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network.
Check for updates
Remember, we're here to help.
Click here to learn about Netgate 24/7/365 support services.
User survey
Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous)
Anonymous User Survey
Useful resources.
Learn more about Netgate's product line, services, and pfSense software from our website
To learn about Netgate appliances and other offers, visit our store
Become part of the pfSense community. Visit our forum
Subscribe to our newsletter for ongoing product information, software announcements and special offers.
Finish
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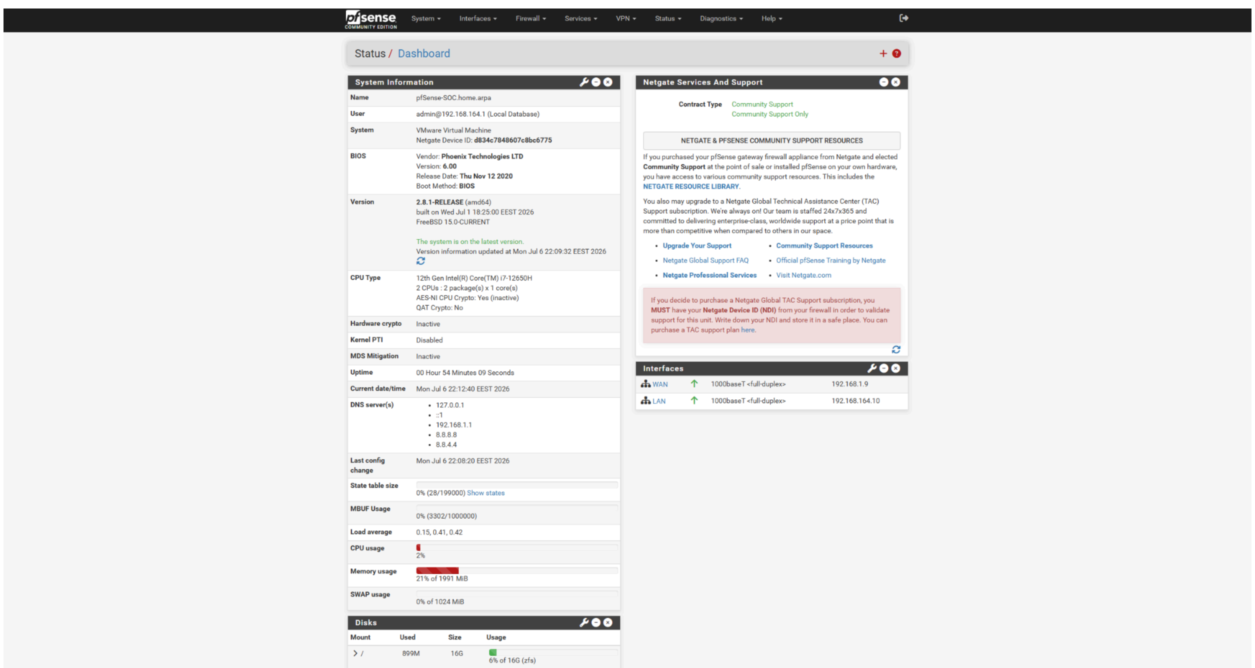
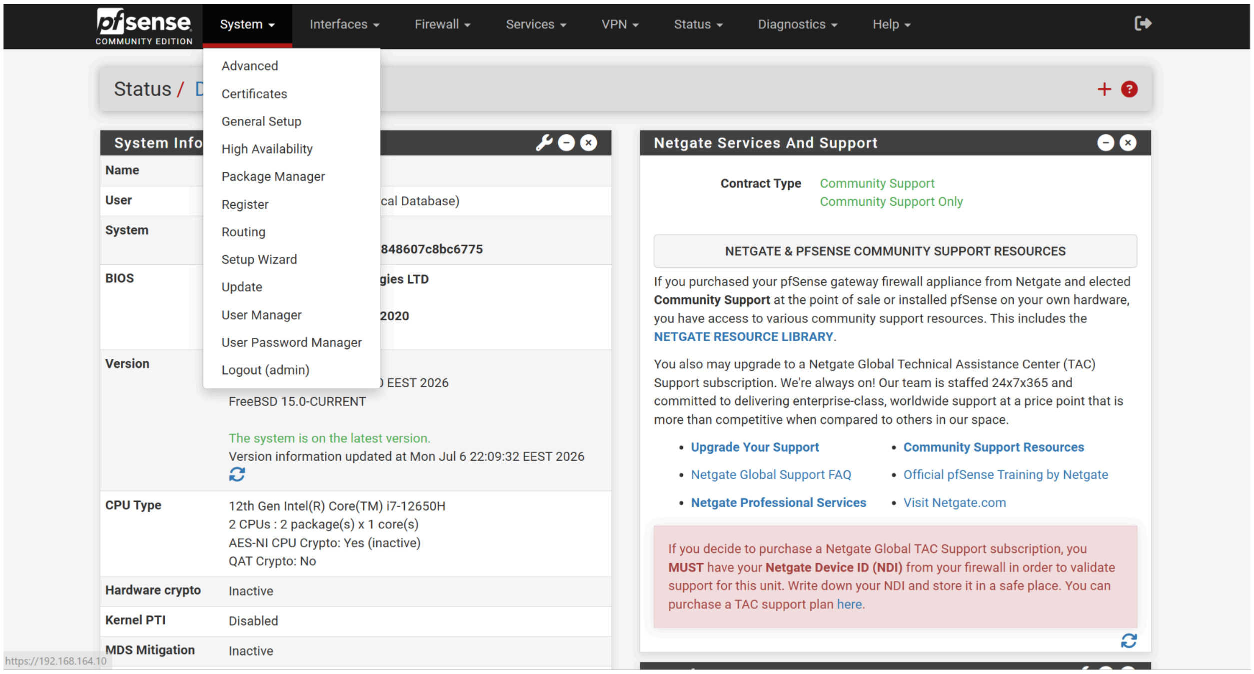
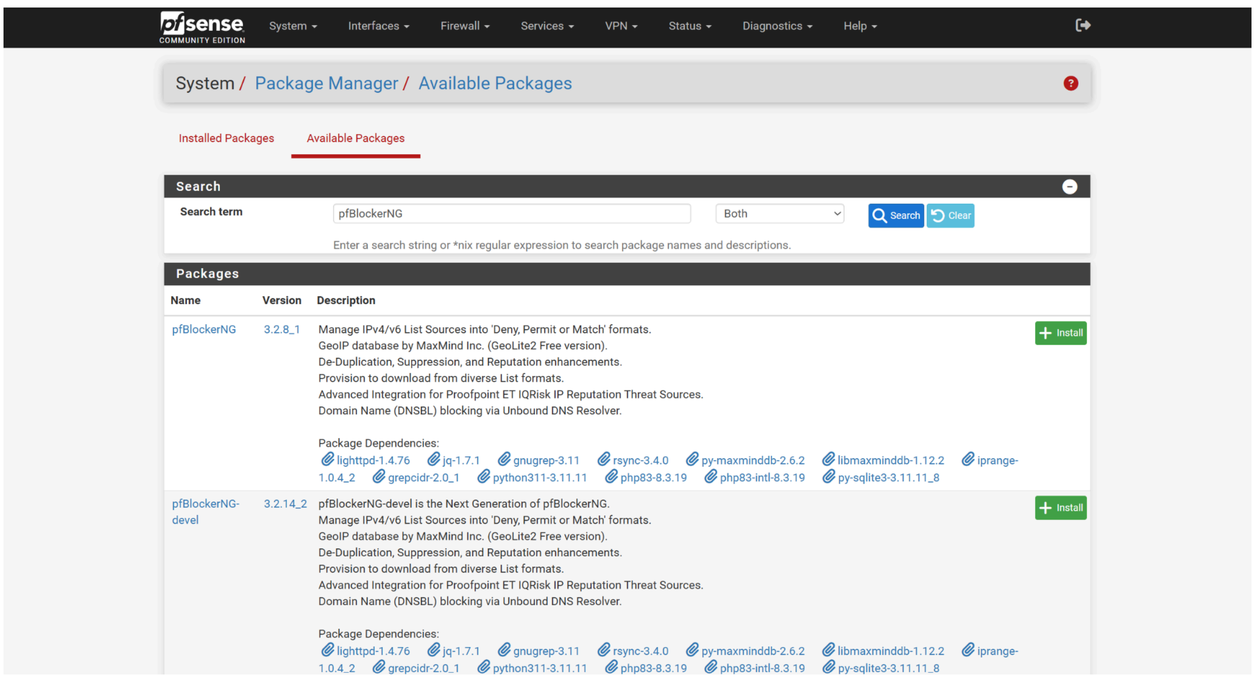
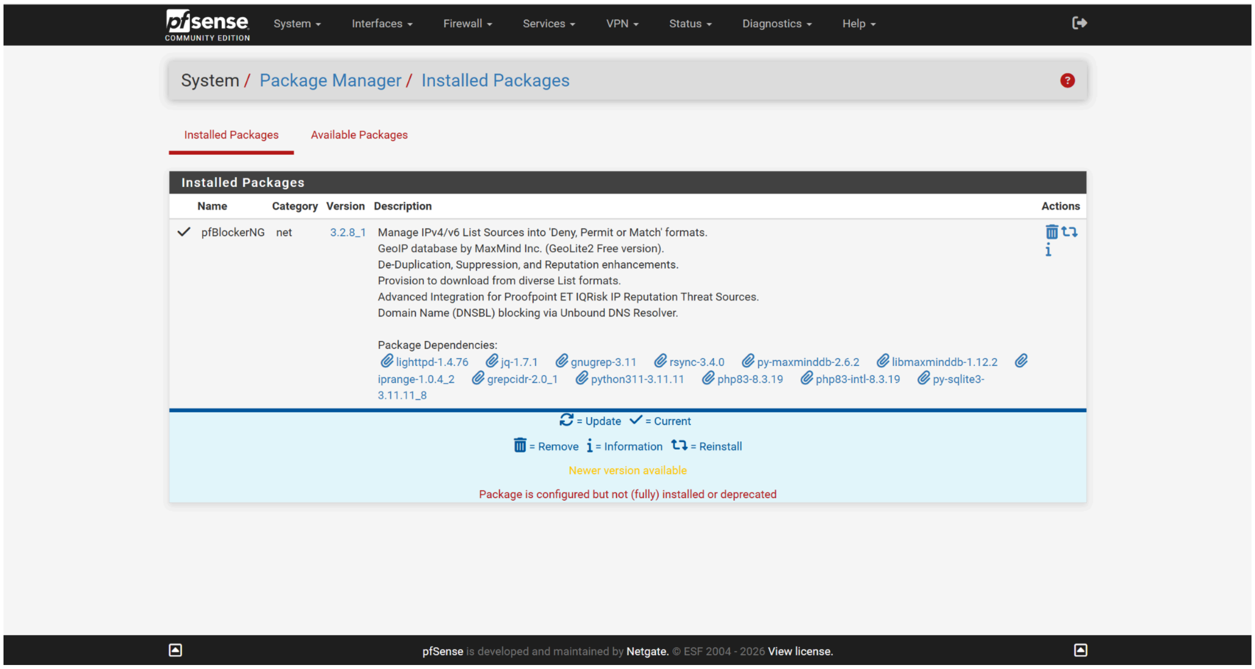

pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Status / Dashboard
System Information
Name pfSense-SOC.home.arpa
User admin@192.168.164.1 (Local Database)
System VMware Virtual Machine
Netgate Device ID: d834c7848607c8bc6775
BIOS Vendor: Phoenix Technologies LTD
Version: 6.00
Release Date: Thu Nov 12 2020
Boot Method: BIOS
Version 2.8.1-RELEASE (amd64)
built on Wed Jul 1 18:25:00 EEST 2026
FreeBSD 15.0-CURRENT
The system is on the latest version.
Version information updated at Mon Jul 6 22:09:32 EEST 2026
CPU Type 12th Gen Intel(R) Core(TM) i7-12650H
2 CPUs : 2 package(s) x 1 core(s)
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Hardware crypto Inactive
Kernel PTI Disabled
MDS Mitigation Inactive
Uptime 00 Hour 54 Minutes 09 Seconds
Current date/time Mon Jul 6 22:12:40 EEST 2026
DNS server(s) 127.0.0.1
::1
192.168.1.1
8.8.8.8
8.8.4.4
Last config change Mon Jul 6 22:08:20 EEST 2026
State table size 0% (28/199000) Show states
MBUF Usage 0% (3302/1000000)
Load average 0.15, 0.41, 0.42
CPU usage 2%
Memory usage 21% of 1991 MiB
SWAP usage 0% of 1024 MiB
Disks
Mount Used Size Usage
/ 899M 16G 6% of 16G (zfs)
Netgate Services And Support
Contract Type Community Support
Community Support Only
NETGATE & PFSENSE COMMUNITY SUPPORT RESOURCES
If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY.
You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.
Upgrade Your Support
Community Support Resources
Netgate Global Support FAQ
Official pfSense Training by Netgate
Netgate Professional Services
Visit Netgate.com
If you decide to purchase a Netgate Global TAC Support subscription, you MUST have your Netgate Device ID (NDI) from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase a TAC support plan here.
Interfaces
WAN 1000baseT <full-duplex> 192.168.1.9
LAN 1000baseT <full-duplex> 192.168.164.10

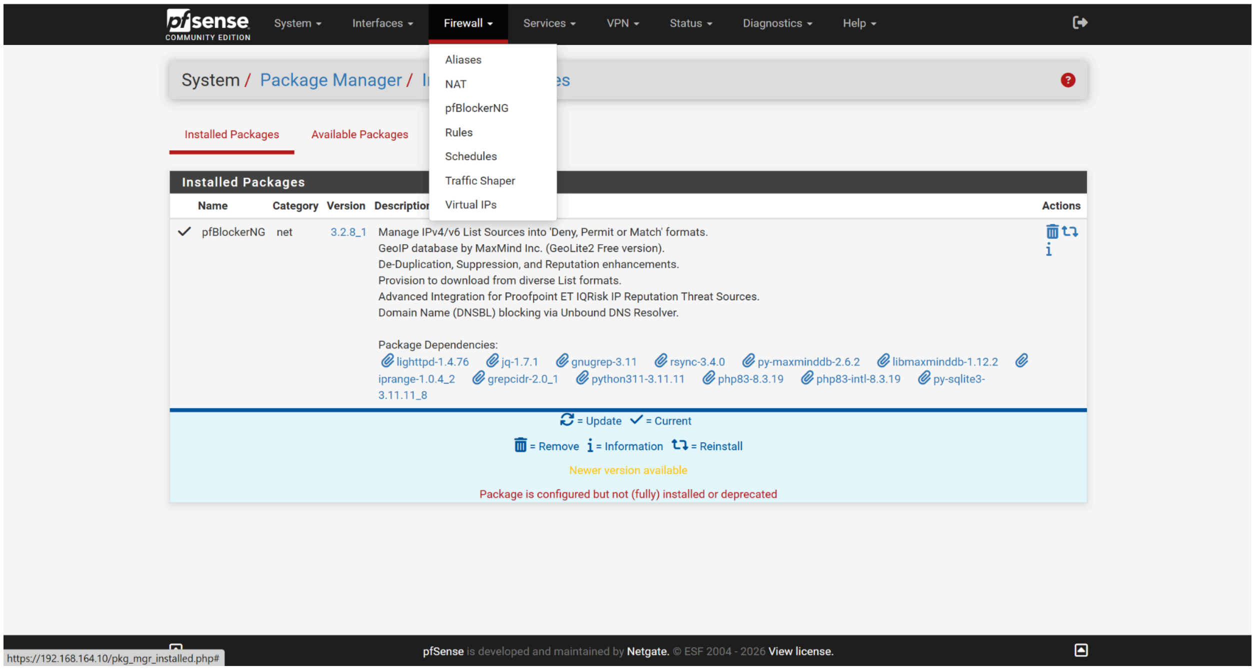
Install the pfBlockerNG Package

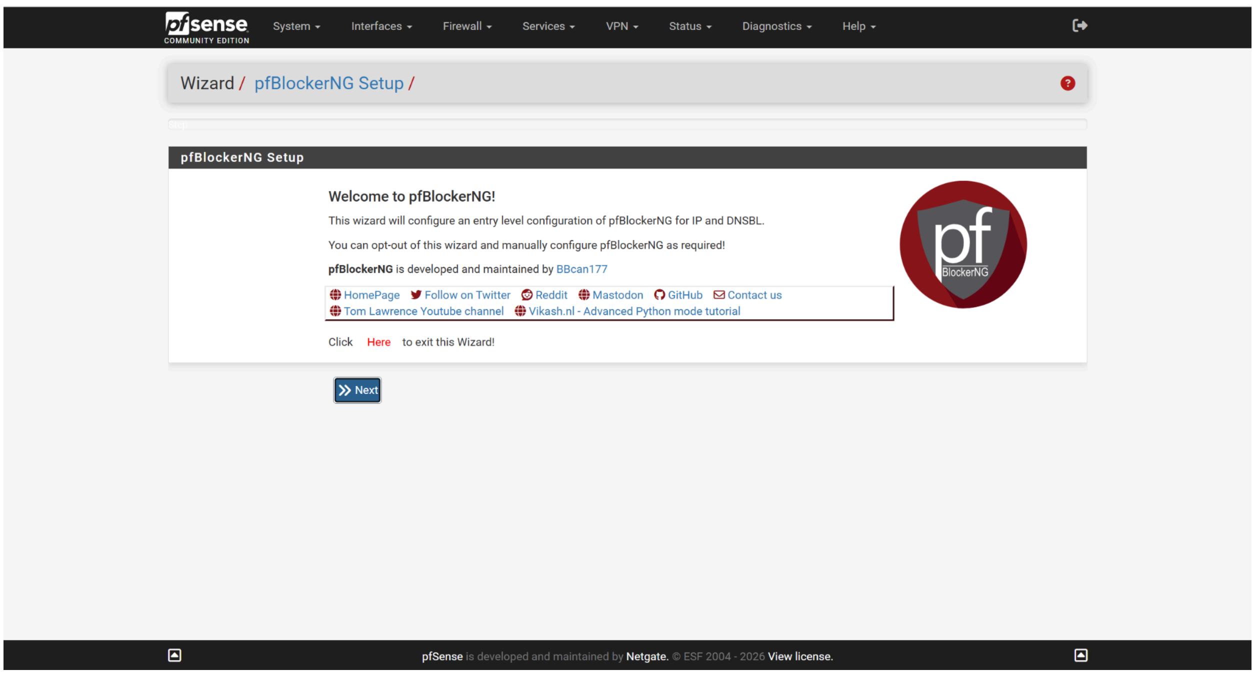
1. Go to System > Package Manager > Available Packages.
2. Search for 'pfBlockerNG' (use pfBlockerNG-devel for the actively maintained branch, recommended).
3. Click Install, confirm the installation, and wait for the package to download and install.
4. After installation, go to Firewall > pfBlockerNG > Wizard to run the initial setup wizard, which creates the default IPv4/IPv6 aliases and DNSBL components.
5. On the General tab, enable pfBlockerNG and keep 'Enable pfBlockerNG DNSBL' checked if you plan to use DNSBL filtering (configured in Step 5).
6. Save and apply the configuration, then force an initial update via Update > Force Update to download the GeoIP and DNSBL data feeds (MaxMind GeoIP database registration may be required — enter a free MaxMind license key under GeoIP tab if prompted).


pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Status /
Advanced
Certificates
General Setup
High Availability
Package Manager
Register
Routing
Setup Wizard
Update
User Manager
User Password Manager
Logout (admin)
System Info
Name
User
cal Database)
System
848607c8bc6775
BIOS
gies LTD
2020
Version
) EEST 2026
FreeBSD 15.0-CURRENT
The system is on the latest version.
Version information updated at Mon Jul 6 22:09:32 EEST 2026
CPU Type
12th Gen Intel(R) Core(TM) i7-12650H
2 CPUs : 2 package(s) x 1 core(s)
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Hardware crypto
Inactive
Kernel PTI
Disabled
MDS Mitigation
Inactive
https://192.168.164.10
Netgate Services And Support
Contract Type
Community Support
Community Support Only
NETGATE & PFSENSE COMMUNITY SUPPORT RESOURCES
If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY.
You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.
Upgrade Your Support
Community Support Resources
Netgate Global Support FAQ
Official pfSense Training by Netgate
Netgate Professional Services
Visit Netgate.com
If you decide to purchase a Netgate Global TAC Support subscription, you MUST have your Netgate Device ID (NDI) from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase a TAC support plan here.


pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
System / Package Manager / Available Packages
Installed Packages
Available Packages
Search
Search term
pfBlockerNG
Both
Search
Clear
Enter a search string or *nix regular expression to search package names and descriptions.
Packages
Name
Version
Description
pfBlockerNG
3.2.8_1
Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats.
GeoIP database by MaxMind Inc. (GeoLite2 Free version).
De-Duplication, Suppression, and Reputation enhancements.
Provision to download from diverse List formats.
Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources.
Domain Name (DNSBL) blocking via Unbound DNS Resolver.
Package Dependencies:
lighttpd-1.4.76 jq-1.7.1 gnugrep-3.11 rsync-3.4.0 py-maxminddb-2.6.2 libmaxminddb-1.12.2 iprange-1.0.4_2 grepcidr-2.0_1 python311-3.11.11 php83-8.3.19 php83-intl-8.3.19 py-sqlite3-3.11.11_8
Install
pfBlockerNG-devel
3.2.14_2
pfBlockerNG-devel is the Next Generation of pfBlockerNG.
Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats.
GeoIP database by MaxMind Inc. (GeoLite2 Free version).
De-Duplication, Suppression, and Reputation enhancements.
Provision to download from diverse List formats.
Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources.
Domain Name (DNSBL) blocking via Unbound DNS Resolver.
Package Dependencies:
lighttpd-1.4.76 jq-1.7.1 gnugrep-3.11 rsync-3.4.0 py-maxminddb-2.6.2 libmaxminddb-1.12.2 iprange-1.0.4_2 grepcidr-2.0_1 python311-3.11.11 php83-8.3.19 php83-intl-8.3.19 py-sqlite3-3.11.11_8
Install


pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
System / Package Manager / Installed Packages
Installed Packages
Available Packages
Installed Packages
Name
Category
Version
Description
Actions
pfBlockerNG
net
3.2.8_1
Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats.
GeoIP database by MaxMind Inc. (GeoLite2 Free version).
De-Duplication, Suppression, and Reputation enhancements.
Provision to download from diverse List formats.
Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources.
Domain Name (DNSBL) blocking via Unbound DNS Resolver.
Package Dependencies:
lighttpd-1.4.76
jq-1.7.1
gnugrep-3.11
rsync-3.4.0
py-maxminddb-2.6.2
libmaxminddb-1.12.2
iprange-1.0.4_2
grepcidr-2.0_1
python311-3.11.11
php83-8.3.19
php83-intl-8.3.19
py-sqlite3-3.11.11_8
= Update
= Current
= Remove
= Information
= Reinstall
Newer version available
Package is configured but not (fully) installed or deprecated
pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.

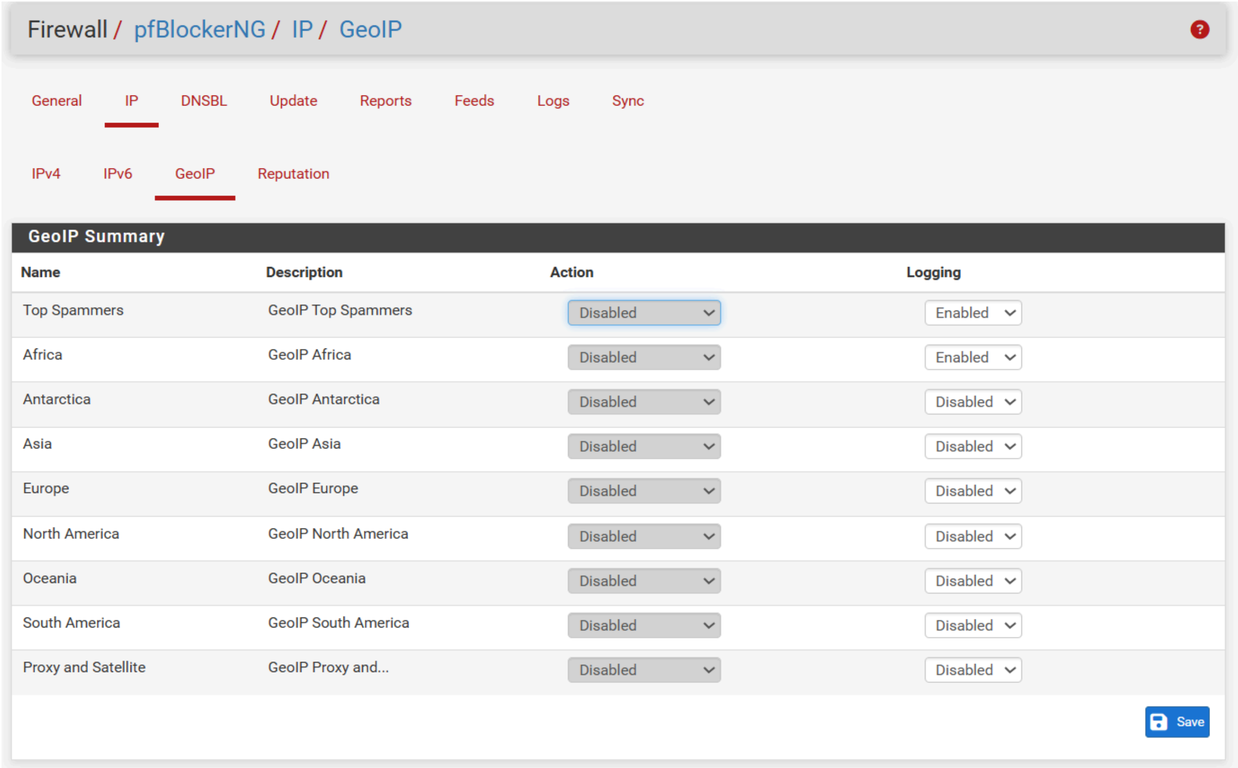

pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Aliases
NAT
pfBlockerNG
Rules
Schedules
Traffic Shaper
Virtual IPs
System / Package Manager /
Installed Packages
Available Packages
Installed Packages
Name
Category
Version
Description
Actions
pfBlockerNG
net
3.2.8_1
Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats.
GeoIP database by MaxMind Inc. (GeoLite2 Free version).
De-Duplication, Suppression, and Reputation enhancements.
Provision to download from diverse List formats.
Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources.
Domain Name (DNSBL) blocking via Unbound DNS Resolver.
Package Dependencies:
lighttpd-1.4.76 jq-1.7.1 gnugrep-3.11 rsync-3.4.0 py-maxminddb-2.6.2 libmaxminddb-1.12.2 iprange-1.0.4_2 grepcidr-2.0_1 python311-3.11.11 php83-8.3.19 php83-intl-8.3.19 py-sqlite3-3.11.11_8
= Update = Current
= Remove = Information = Reinstall
Newer version available
Package is configured but not (fully) installed or deprecated
https://192.168.164.10/pkg_mgr_installed.php#
pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.

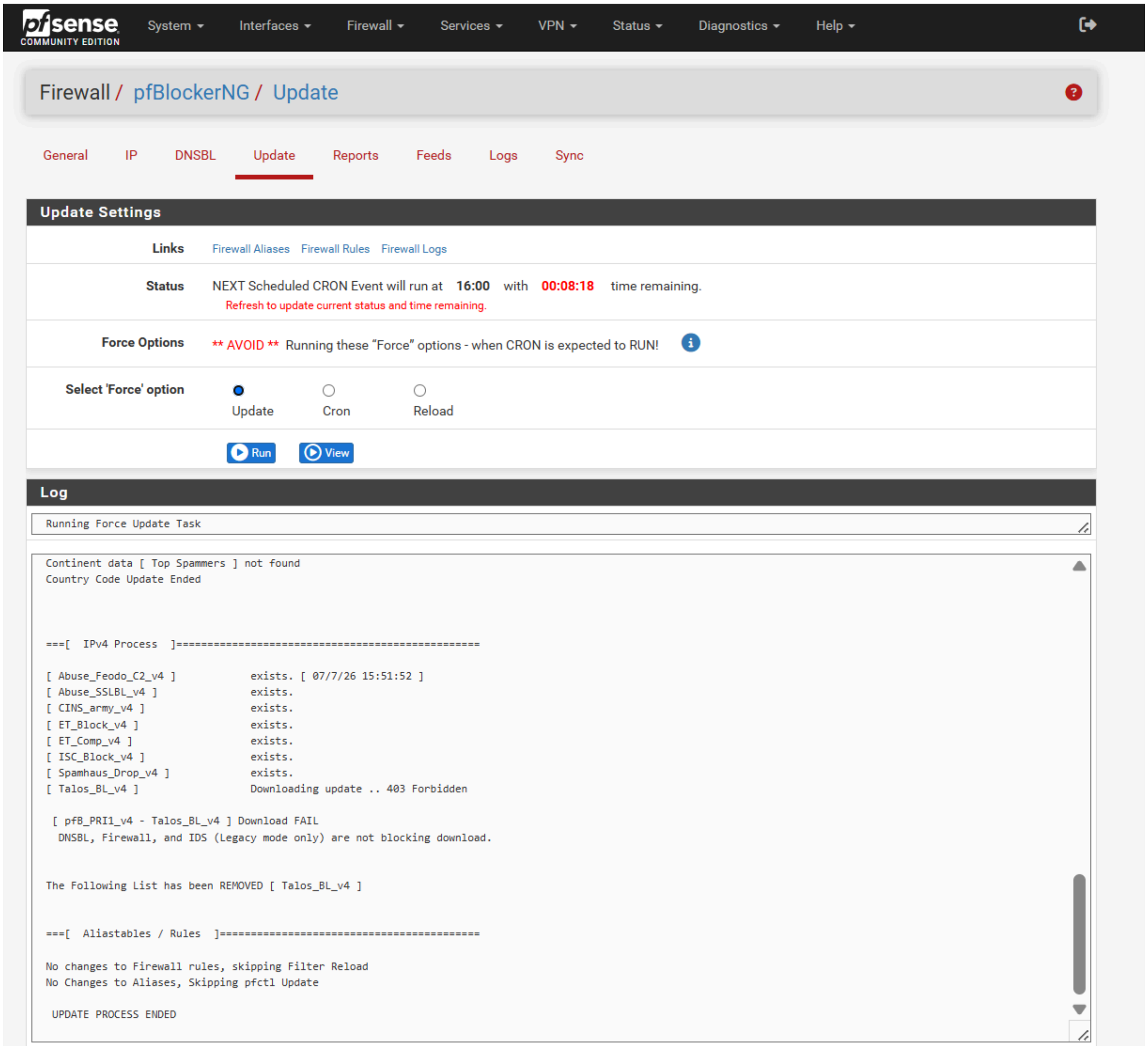

pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Wizard / pfBlockerNG Setup /
pfBlockerNG Setup
Welcome to pfBlockerNG!
This wizard will configure an entry level configuration of pfBlockerNG for IP and DNSBL.
You can opt-out of this wizard and manually configure pfBlockerNG as required!
pfBlockerNG is developed and maintained by BBcan177
HomePage Follow on Twitter Reddit Mastodon GitHub Contact us
Tom Lawrence Youtube channel Vikash.nl - Advanced Python mode tutorial
Click Here to exit this Wizard!
pf BlockerNG
Next
pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.

Configure GeoIP-Based Country Blocking

GeoIP blocking uses MaxMind country databases to build firewall aliases that group all IP ranges belonging to a selected country, which are then referenced in a firewall rule to deny traffic.

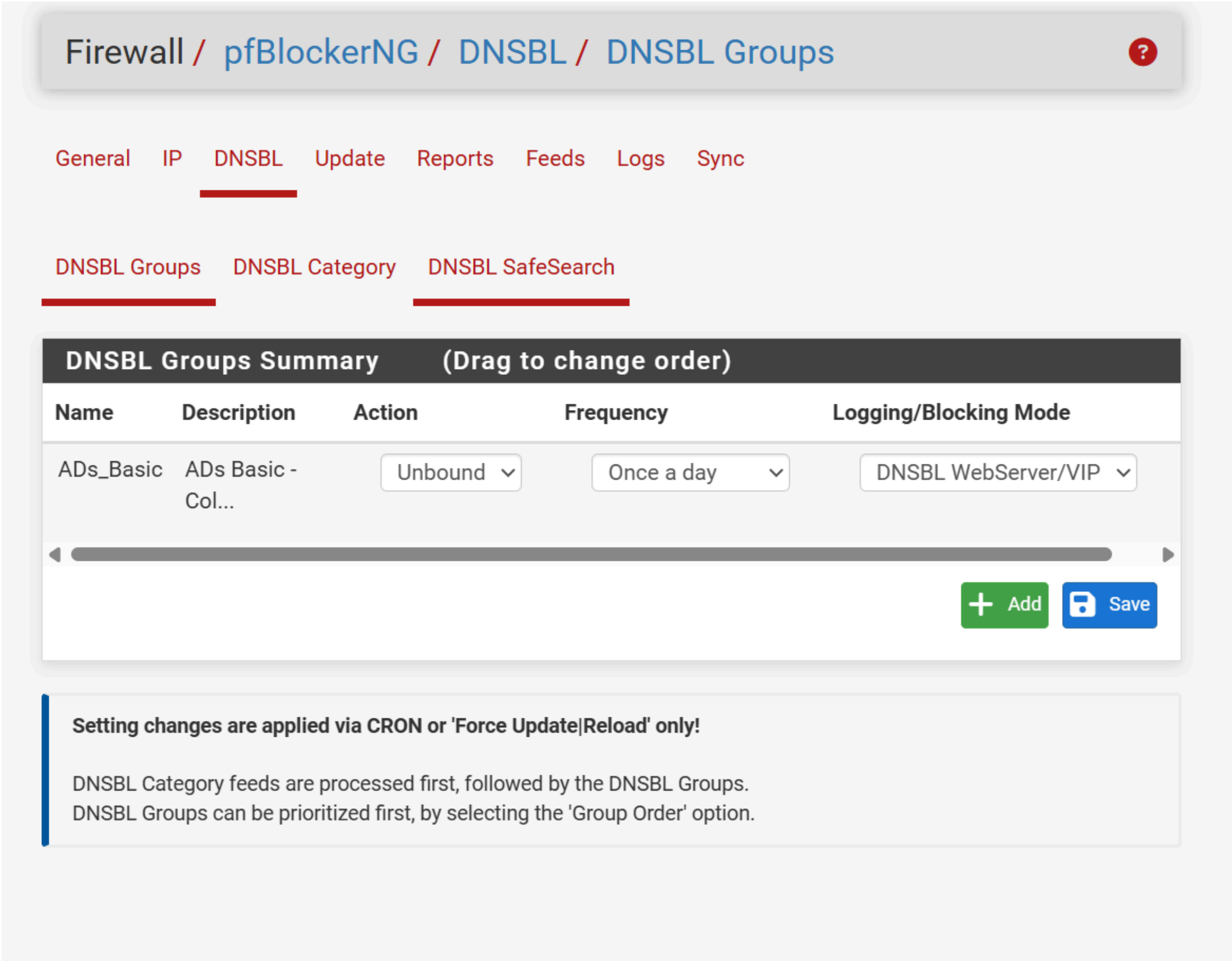
1. Navigate to Firewall > pfBlockerNG > IP tab.
2. Under the IPv4 section, click the green + to add a new GeoIP source, or use the pre-built 'GeoIP' category.
3. Select the target country to block (e.g. Netherlands — NL), or add multiple countries as required by the assignment (e.g. high-risk countries list).
4. Set the Action to 'Deny Both' (blocks inbound and outbound traffic to/from the selected country) and enable Auto and Cron for automatic list updates.
5. Save the settings, then go to Update > Force Update to build the alias (this generates a pfB_NL or similarly named firewall alias).
6. Go to Firewall > Rules > WAN and confirm that pfBlockerNG has automatically inserted a deny rule referencing the new GeoIP alias at the top of the ruleset (pfBlockerNG auto-creates rules when 'Auto' is enabled).

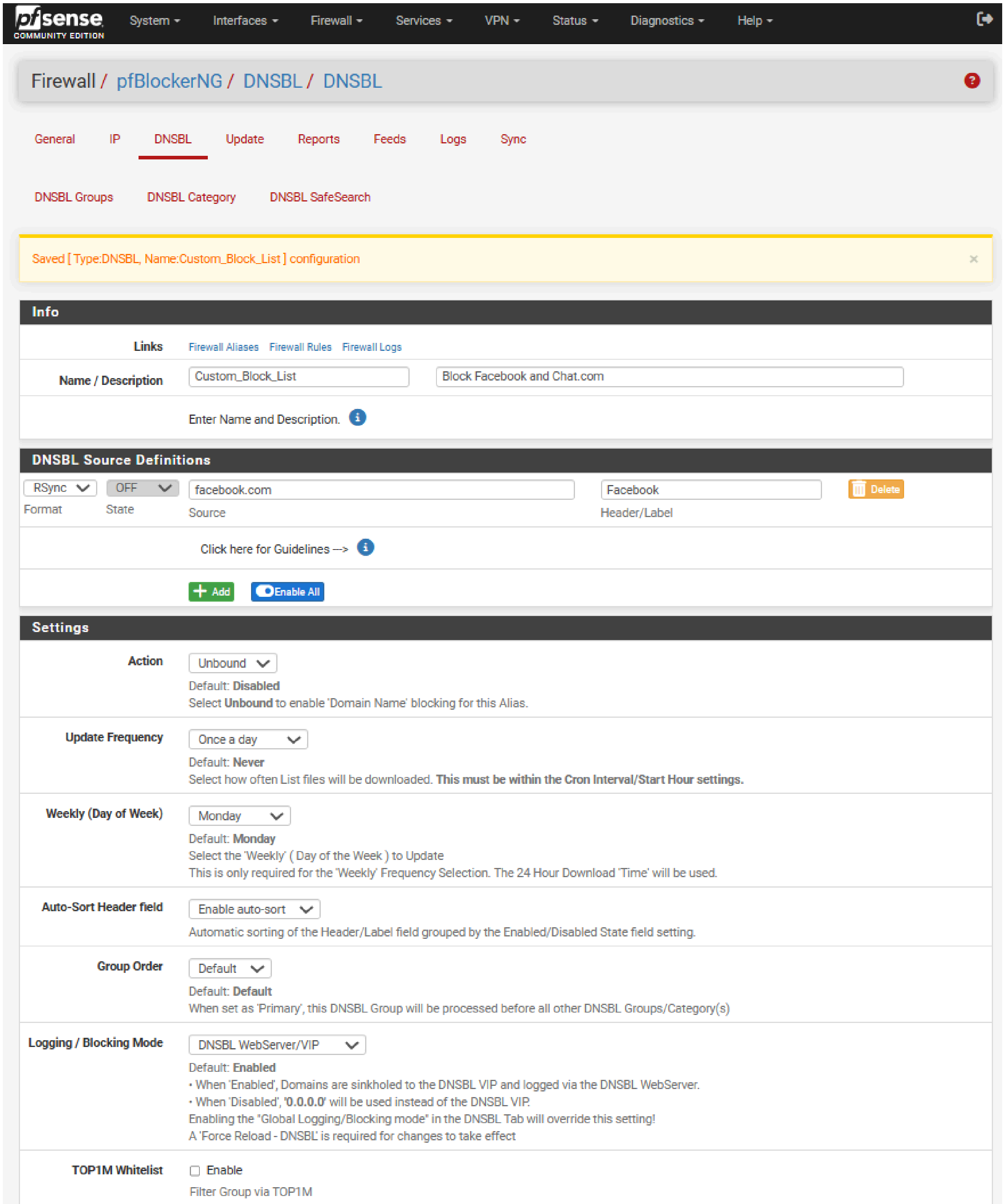

Firewall / pfBlockerNG / IP / GeoIP
General
IP
DNSBL
Update
Reports
Feeds
Logs
Sync
IPv4
IPv6
GeoIP
Reputation
GeoIP Summary
Name
Description
Action
Logging
Top Spammers
GeoIP Top Spammers
Disabled
Enabled
Africa
GeoIP Africa
Disabled
Enabled
Antarctica
GeoIP Antarctica
Disabled
Disabled
Asia
GeoIP Asia
Disabled
Disabled
Europe
GeoIP Europe
Disabled
Disabled
North America
GeoIP North America
Disabled
Disabled
Oceania
GeoIP Oceania
Disabled
Disabled
South America
GeoIP South America
Disabled
Disabled
Proxy and Satellite
GeoIP Proxy and...
Disabled
Disabled
Save

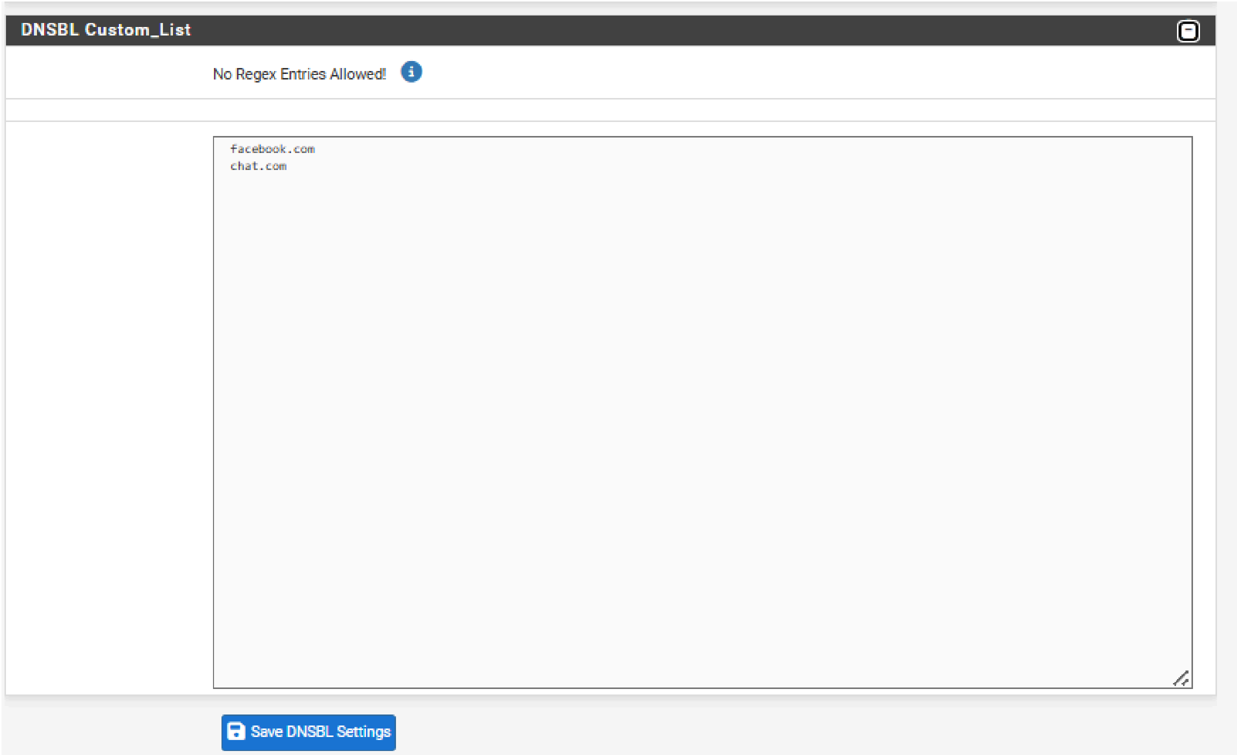


DNSBL Website Blocking Test

From the LAN test client, attempt to browse to facebook.com and chat.com. Confirm the DNSBL block page is returned instead of the real site, and verify the block is logged under Firewall > pfBlockerNG > DNSBL Stats/Reports.


Firewall / pfBlockerNG / DNSBL / DNSBL Groups
General
IP
DNSBL
Update
Reports
Feeds
Logs
Sync
DNSBL Groups
DNSBL Category
DNSBL SafeSearch
DNSBL Groups Summary (Drag to change order)
Name
Description
Action
Frequency
Logging/Blocking Mode
ADs_Basic
ADs Basic - Col...
Unbound
Once a day
DNSBL WebServer/VIP
Add
Save
Setting changes are applied via CRON or 'Force Update|Reload' only!
DNSBL Category feeds are processed first, followed by the DNSBL Groups.
DNSBL Groups can be prioritized first, by selecting the 'Group Order' option.


pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Firewall / pfBlockerNG / DNSBL / DNSBL
General
IP
DNSBL
Update
Reports
Feeds
Logs
Sync
DNSBL Groups
DNSBL Category
DNSBL SafeSearch
Saved [Type:DNSBL, Name:Custom_Block_List] configuration
Info
Links
Firewall Aliases
Firewall Rules
Firewall Logs
Name / Description
Custom_Block_List
Block Facebook and Chat.com
Enter Name and Description.
DNSBL Source Definitions
RSync
OFF
facebook.com
Facebook
Delete
Format
State
Source
Header/Label
Click here for Guidelines -->
Add
Enable All
Settings
Action
Unbound
Default: Disabled
Select Unbound to enable 'Domain Name' blocking for this Alias.
Update Frequency
Once a day
Default: Never
Select how often List files will be downloaded. This must be within the Cron Interval/Start Hour settings.
Weekly (Day of Week)
Monday
Default: Monday
Select the 'Weekly' (Day of the Week) to Update
This is only required for the 'Weekly' Frequency Selection. The 24 Hour Download 'Time' will be used.
Auto-Sort Header field
Enable auto-sort
Automatic sorting of the Header/Label field grouped by the Enabled/Disabled State field setting.
Group Order
Default
Default: Default
When set as 'Primary', this DNSBL Group will be processed before all other DNSBL Groups/Category(s)
Logging / Blocking Mode
DNSBL WebServer/VIP
Default: Enabled
• When 'Enabled', Domains are sinkholed to the DNSBL VIP and logged via the DNSBL WebServer.
• When 'Disabled', '0.0.0.0' will be used instead of the DNSBL VIP.
Enabling the "Global Logging/Blocking mode" in the DNSBL Tab will override this setting!
A 'Force Reload - DNSBL' is required for changes to take effect
TOP1M Whitelist
Enable
Filter Group via TOP1M


DNSBL Custom_List
No Regex Entries Allowed!
facebook.com
chat.com
Save DNSBL Settings

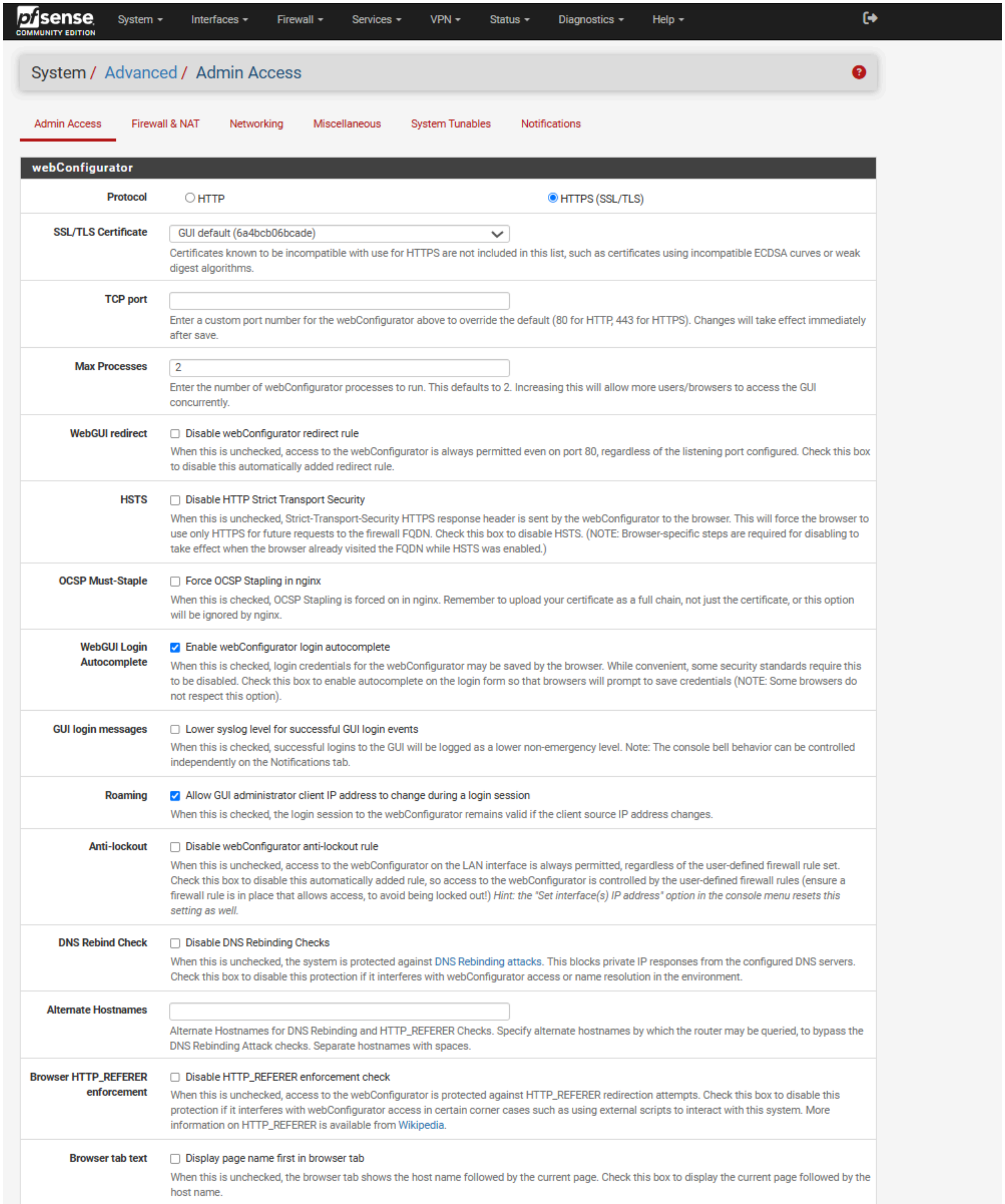

C:\Users\BASMALA>nslookup facebook.com 192.168.164.10
Server: pfSense-SOC.home.arpa
Address: 192.168.164.10
Name: facebook.com
Address: 10.10.10.1
C:\Users\BASMALA>

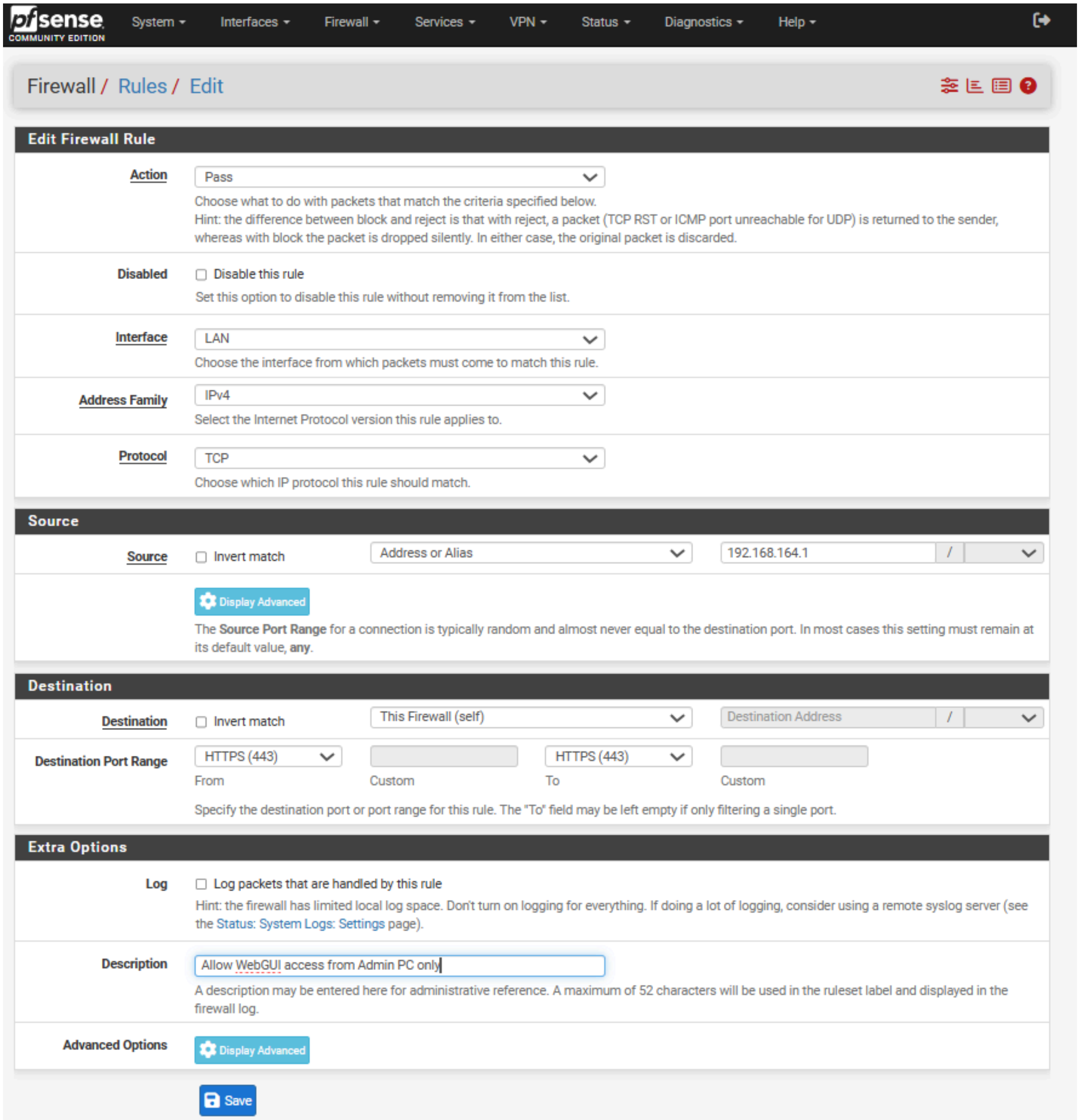

C:\Users\BASMALA>nslookup chat.com 192.168.164.10
Server: pfSense-SOC.home.arpa
Address: 192.168.164.10
Name: chat.com
Address: 10.10.10.1

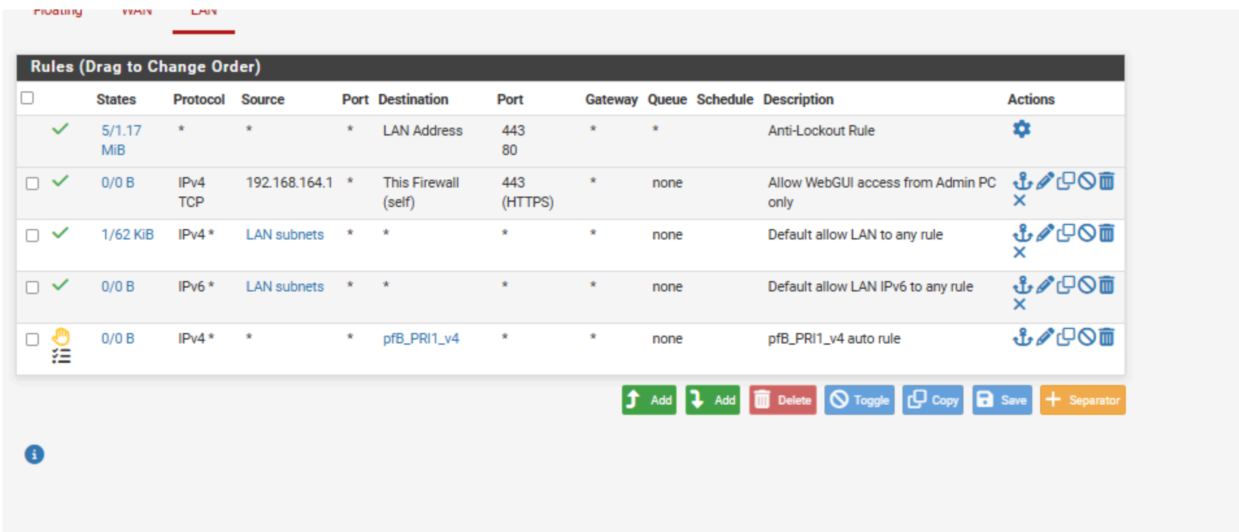
Configure Administrator Access Control

Restricting WebGUI access reduces the attack surface of the firewall's management plane, ensuring only authorized administrators/hosts can reach it.

1. Go to System > Advanced > Admin Access and confirm the WebGUI protocol is set to HTTPS only (disable HTTP redirect if not required).
2. Optionally change the WebGUI port from the default 443 to a non-standard port to reduce automated scanning exposure.
3. Create a dedicated firewall alias under Firewall > Aliases (e.g. 'Admin_Hosts') containing only the trusted management IP address(es) (e.g. 192.168.1.10).
4. Go to Firewall > Rules > LAN and create a rule that allows HTTPS (or the custom WebGUI port) to the firewall's LAN address only from the Admin_Hosts alias, placed above any broader allow rules.
5. Add a final deny rule immediately below it that blocks all other traffic destined to the firewall's own LAN IP on the WebGUI port, preventing any other LAN host from reaching the management interface.
6. Under System > User Manager, review the local admin account, enforce a strong password policy, and optionally enable two-factor authentication for the admin account.
7. Save, apply the changes, and validate from a non-authorized host that the WebGUI is no longer reachable, while the authorized admin host retains access.


pfsense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
System / Advanced / Admin Access
Admin Access
Firewall & NAT
Networking
Miscellaneous
System Tunables
Notifications
webConfigurator
Protocol
HTTP
HTTPS (SSL/TLS)
SSL/TLS Certificate
GUI default (6a4bcb06bcade)
Certificates known to be incompatible with use for HTTPS are not included in this list, such as certificates using incompatible ECDSA curves or weak digest algorithms.
TCP port
Enter a custom port number for the webConfigurator above to override the default (80 for HTTP, 443 for HTTPS). Changes will take effect immediately after save.
Max Processes
2
Enter the number of webConfigurator processes to run. This defaults to 2. Increasing this will allow more users/browsers to access the GUI concurrently.
WebGUI redirect
Disable webConfigurator redirect rule
When this is unchecked, access to the webConfigurator is always permitted even on port 80, regardless of the listening port configured. Check this box to disable this automatically added redirect rule.
HSTS
Disable HTTP Strict Transport Security
When this is unchecked, Strict-Transport-Security HTTPS response header is sent by the webConfigurator to the browser. This will force the browser to use only HTTPS for future requests to the firewall FQDN. Check this box to disable HSTS. (NOTE: Browser-specific steps are required for disabling to take effect when the browser already visited the FQDN while HSTS was enabled.)
OCSP Must-Staple
Force OCSP Stapling in nginx
When this is checked, OCSP Stapling is forced on in nginx. Remember to upload your certificate as a full chain, not just the certificate, or this option will be ignored by nginx.
WebGUI Login Autocomplete
Enable webConfigurator login autocomplete
When this is checked, login credentials for the webConfigurator may be saved by the browser. While convenient, some security standards require this to be disabled. Check this box to enable autocomplete on the login form so that browsers will prompt to save credentials (NOTE: Some browsers do not respect this option).
GUI login messages
Lower syslog level for successful GUI login events
When this is checked, successful logins to the GUI will be logged as a lower non-emergency level. Note: The console bell behavior can be controlled independently on the Notifications tab.
Roaming
Allow GUI administrator client IP address to change during a login session
When this is checked, the login session to the webConfigurator remains valid if the client source IP address changes.
Anti-lockout
Disable webConfigurator anti-lockout rule
When this is unchecked, access to the webConfigurator on the LAN interface is always permitted, regardless of the user-defined firewall rule set. Check this box to disable this automatically added rule, so access to the webConfigurator is controlled by the user-defined firewall rules (ensure a firewall rule is in place that allows access, to avoid being locked out!) Hint: the "Set interface(s) IP address" option in the console menu resets this setting as well.
DNS Rebind Check
Disable DNS Rebinding Checks
When this is unchecked, the system is protected against DNS Rebinding attacks. This blocks private IP responses from the configured DNS servers. Check this box to disable this protection if it interferes with webConfigurator access or name resolution in the environment.
Alternate Hostnames
Alternate Hostnames for DNS Rebinding and HTTP_REFERER Checks. Specify alternate hostnames by which the router may be queried, to bypass the DNS Rebinding Attack checks. Separate hostnames with spaces.
Browser HTTP_REFERER enforcement
Disable HTTP_REFERER enforcement check
When this is unchecked, access to the webConfigurator is protected against HTTP_REFERER redirection attempts. Check this box to disable this protection if it interferes with webConfigurator access in certain corner cases such as using external scripts to interact with this system. More information on HTTP_REFERER is available from Wikipedia.
Browser tab text
Display page name first in browser tab
When this is unchecked, the browser tab shows the host name followed by the current page. Check this box to display the current page followed by the host name.

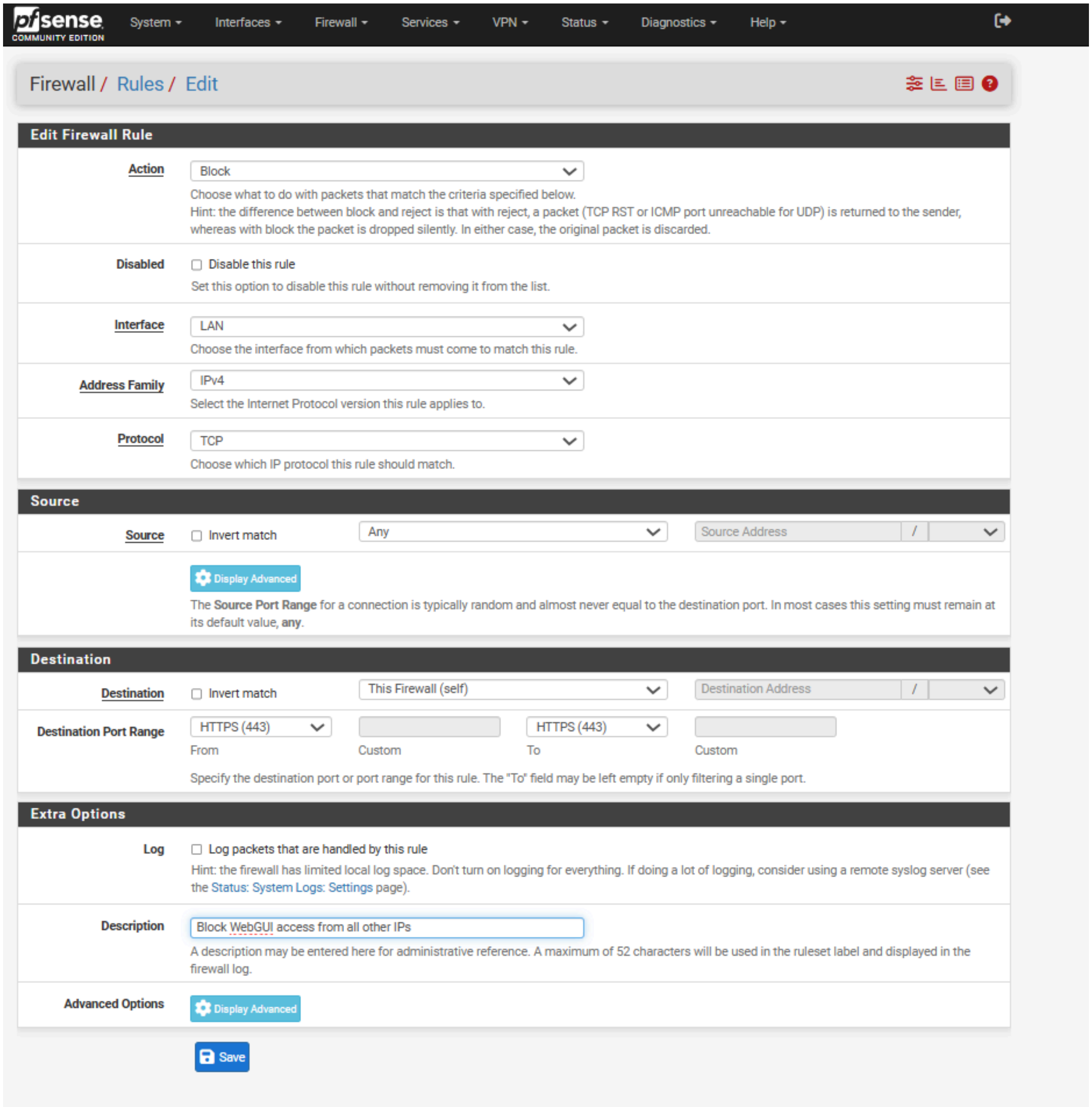

pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
LAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which IP protocol this rule should match.
Source
Source
Invert match
Address or Alias
192.168.164.1
/
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
This Firewall (self)
Destination Address
/
Destination Port Range
HTTPS (443)
From
Custom
HTTPS (443)
To
Custom
Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.
Extra Options
Log
Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).
Description
Allow WebGUI access from Admin PC only
A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.
Advanced Options
Display Advanced
Save

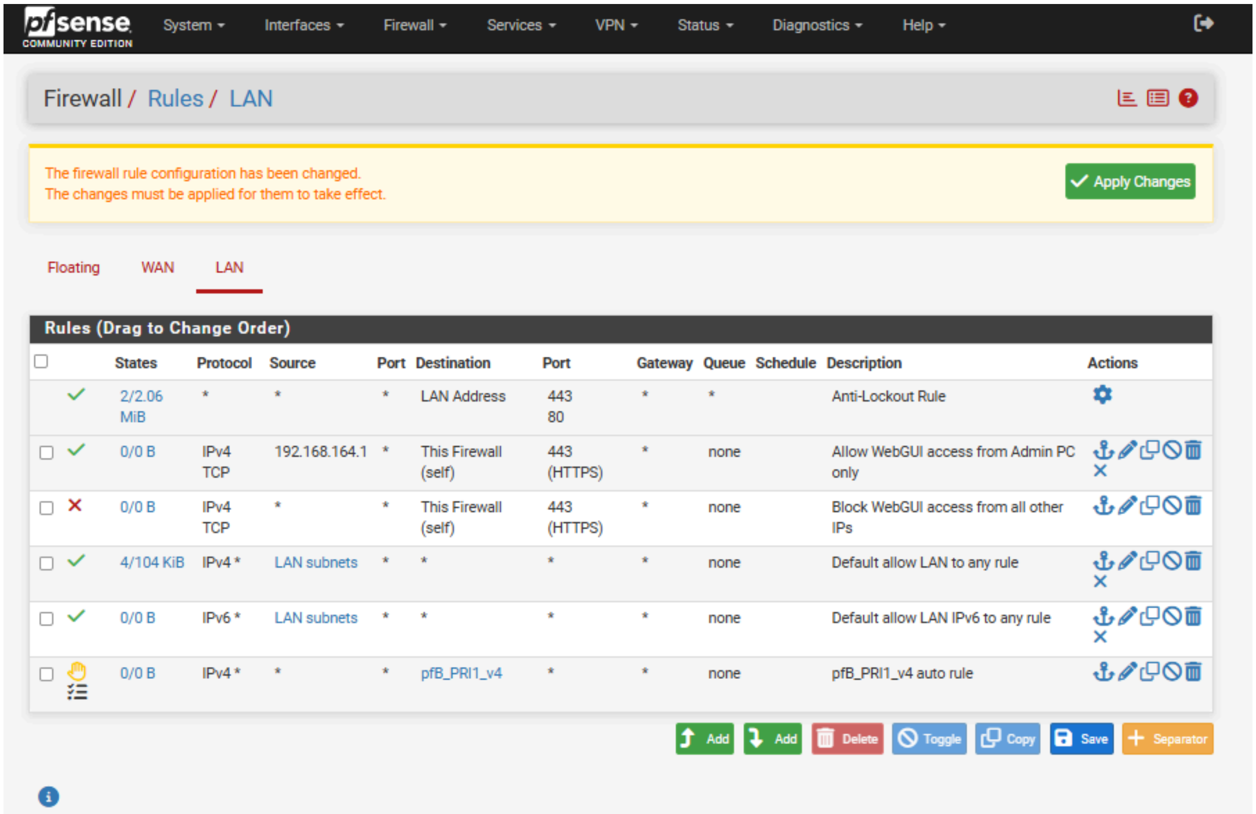


Rules (Drag to Change Order)

		States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	✓	5/1.17 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
	✓	0/0 B	IPv4 TCP	192.168.164.1	*	This Firewall (self)	443 (HTTPS)	*	none		Allow WebGUI access from Admin PC only	
	✓	1/62 KiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
	✓	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	
		0/0 B	IPv4 *	*	*	pfB_PRI1_v4	*	*	none		pfB_PRI1_v4 auto rule	

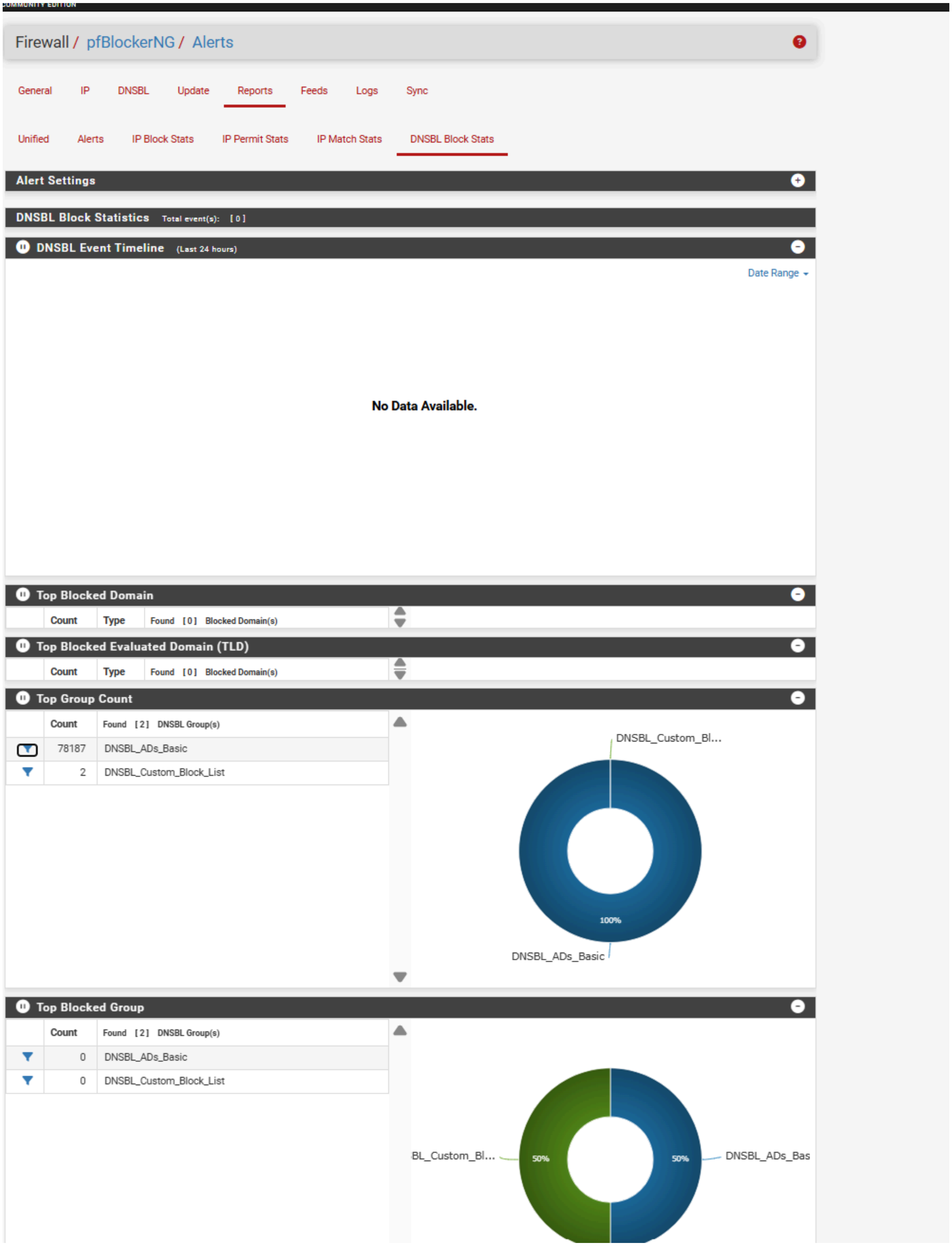
Add Add Delete Toggle Copy Save Separator


pfsense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Firewall / Rules / Edit
Edit Firewall Rule
Action
Block
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
LAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which IP protocol this rule should match.
Source
Source
Invert match
Any
Source Address
/
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
This Firewall (self)
Destination Address
/
Destination Port Range
HTTPS (443)
From
Custom
HTTPS (443)
To
Custom
Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.
Extra Options
Log
Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).
Description
Block WebGUI access from all other IPs
A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.
Advanced Options
Display Advanced
Save



Monitor & Validate Firewall Activity

1. Navigate to Status > System Logs > Firewall to view real-time blocked/allowed connections, and filter by the pfBlockerNG-generated rule labels (e.g. rule descriptions containing the country code or DNSBL group name).
2. Navigate to Firewall > pfBlockerNG > Reports (or the Alerts tab, depending on version) to view aggregated statistics: top blocked countries, top blocked IPs, and DNSBL query/block counts.
3. Cross-reference a blocked connection attempt in the System Logs with the corresponding entry in the pfBlockerNG Reports to confirm both are logging the same event consistently.
4. Optionally configure Status > System Logs > Settings to increase log retention or forward logs to a remote syslog server for centralized SOC monitoring.



COMMUNITY EDITION
Firewall / pfBlockerNG / Alerts
General
IP
DNSBL
Update
Reports
Feeds
Logs
Sync
Unified
Alerts
IP Block Stats
IP Permit Stats
IP Match Stats
DNSBL Block Stats
Alert Settings
DNSBL Block Statistics Total event(s): [0]
DNSBL Event Timeline (Last 24 hours)
Date Range
No Data Available.
Top Blocked Domain
Count Type Found [0] Blocked Domain(s)
Top Blocked Evaluated Domain (TLD)
Count Type Found [0] Blocked Domain(s)
Top Group Count
Count Found [2] DNSBL Group(s)
78187 DNSBL_ADs_Basic
2 DNSBL_Custom_Block_List
DNSBL_Custom_Bl...
100%
DNSBL_ADs_Basic
Top Blocked Group
Count Found [2] DNSBL Group(s)
0 DNSBL_ADs_Basic
0 DNSBL_Custom_Block_List
BL_Custom_Bl...
50%
50%
DNSBL_ADs_Bas

Conclusion

This lab successfully demonstrated deployment of pfSense as a virtual firewall appliance and the configuration of layered security controls using pfBlockerNG. GeoIP-based country blocking reduced exposure to traffic from unwanted geographic regions, DNSBL filtering enforced acceptable-use policy by blocking specific websites at the DNS layer, and administrative access control hardened the firewall's management plane against unauthorized access. Ongoing monitoring through pfBlockerNG reports and the firewall's system logs confirmed that all configured policies were actively enforced and generating verifiable security events. These skills form a practical foundation for Blue Team responsibilities including perimeter defense, policy enforcement, and continuous security monitoring.